

FPGA-based SoC Design

International Master of Science in Electrical Engineering

Prof. Dr. C. Jakob

University of Applied Sciences Darmstadt

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Faculty of Electrical Engineering and Information Technology

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Course Organization and Introduction

Today's Agenda

Intended topics for today's session

- General Course Information
- Lecture Preview
- Course Goals and Intended Learning Outcomes
- Prerequisites
- Exam and Evaluation
- Recommended Readings
- FSoC WS22/23 in a Nutshell

General Course Information

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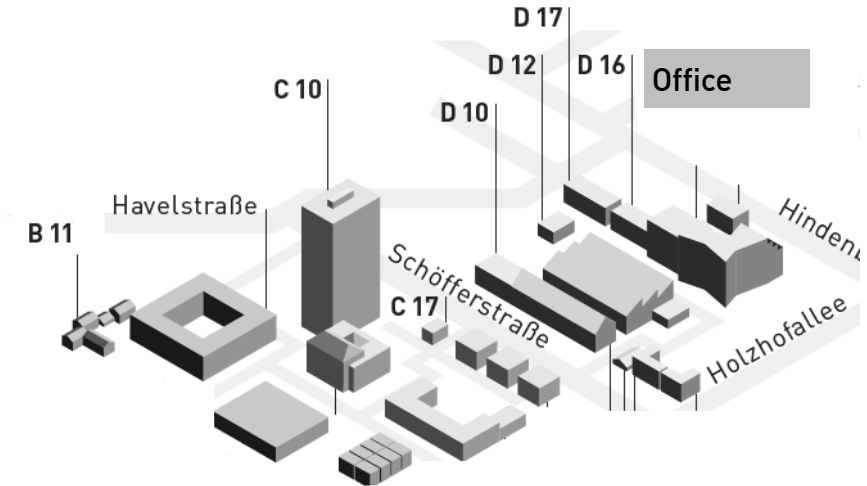
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General Course Information

h_da – fbeit - FPGA-based SoC Design

Contact Information

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- Office: 311/D16, Birkenweg 8, 64295 Darmstadt
- Email: christian.jakob@h-da.de – Use your official h_da address to contact me
- Phone: +49.6151.16-37706 (Rarely reachable by phone)
- Office hour: Tuesday, 9.00am (this might change again)
no registration necessary, first come
first serve ...



General Course Information

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Moodle

- Moodle is the main landing page
- Course notes, additional materials, assignments, readings, example code ...
... will be posted on moodle
- Use the moodle forum to ask your questions. Something that is not clear to you is most probably not clear to others as well!

Lecture Preview

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Intended Roadmap WS22/23

- **Design and Verification of FPGA based System using SystemVerilog**

Intel based FPGA Design, SystemVerilog based Design Entry, Build Automation using Makefiles, ...

- **The Architecture of modern FPGAs – How to map HDL to HW ...**

Fundamental building blocks of modern FPGAs (Logic cells, memory, DSP cells, PLLs, ...) and how to efficiently use them in HDL

- **Designing Data and Control-Path Architectures using SystemVerilog**

Designing Data and Control Path Architectures, State Machines, ...

- **Designing FPGA based Softcore Processor Systems**

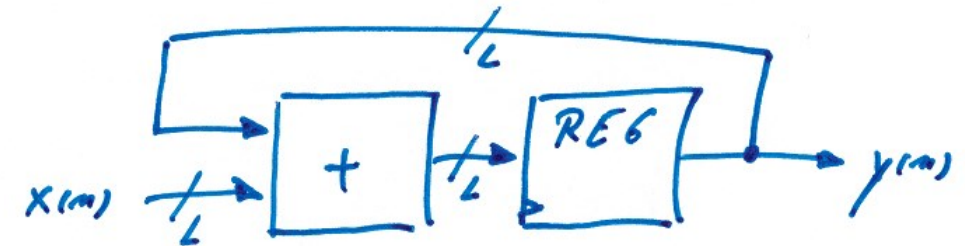
Processor Design and Programming of FPGA based Softcore Architectures

- **Embedded Linux and SoC FPGAs**

Architecture of SoC FPGAs, core concepts – from booting up to writing user space applications

- **Design and Interfacing of Custom Accelerators**

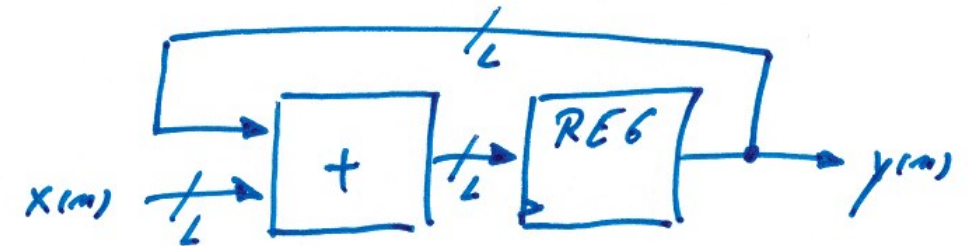
Implementing custom Accelerators, ISA Customization and Extensions, Design of Customized HW/SW Interfaces



Intended Roadmap WS22/23

- **High-Level-Synthesis**

Algorithm and interface synthesis, design evaluation and optimization



Course Goals and Intended Learning Outcomes

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Course Goals and Learning Outcomes

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Intended Learning Outcomes

Students should understand

- the structure of SoC FPGAs and their role in the design of modern electronic systems.
- the tools and methodologies for FPGA-based SoC design.
- the basic principles of hardware/software co-design and co-verification.

Course Goals and Learning Outcomes

h_da – fbeit - FPGA-based SoC Design

Intended Learning Outcomes

Students should be able to apply the gained knowledge

- to partition simple software programs into dedicated hard- and software components.
- to analyse existing HW/SW architectures, to identify possible performance bottlenecks and to optimize them by finding the optimal HW/SW mapping.
- to optimize existing HW/SW architectures by transformations on hardware and software components.
- to evaluate the implementation results (e.g. timing, resource usage, power consumption) and correlate them with the corresponding high level design.

Prerequisites

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Prerequisites

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Prerequisites

- I assume that students are familiar with the concepts of “hardware oriented” C programming and the fundamentals of digital logic design.
- Catch up on untreated material as soon as possible.

Laboratory Sessions

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FSoC Lab WS22/23

- In order to participate in the module exam, it is required to successfully finish the lab part of the module.
- Successful fulfilment of prerequisites are measured by:
 - Lab attendance
 - Lab entry test
 - Lab progress
- 25% of the module grade is obtained by the laboratory.
- There are four lab sessions in total.
- Lab seat assignments will be published shortly.

FSoC Lab WS22/23

- The lab focuses on teaching practical skills related to FPGA based SoC design using C and SystemVerilog:
 - Design and implementation of custom hardware accelerators and ISA extensions.
 - HW/SW integration of custom accelerators into existing FPGA based SoC architectures followed by profiling and benchmarking of the respective solutions.

FSoC Lab WS22/23 – Hardware Accelerator Design

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FSoC Lab WS22/23 – Hardware Accelerator Design

- General-purpose CPUs implement a fixed set of instructions, able to execute nearly any kind of computation.
- This great flexibility offered by a CPU based computing platform also represents one of its major drawbacks.
- The sequential execution of algorithms, often characterized as ‘temporal computing’ allows the implementation of almost all existing algorithms, however, not always with their best potential performance.

FSoC Lab WS22/23 – Hardware Accelerator Design

- **Wikipedia:** Hardware acceleration is the use of computer hardware designed to perform specific functions more efficiently when compared to software running on a general-purpose central processing unit (CPU).
- The slowdown and the nearing end of transistor scaling play a vital role in the rise of domain-specific architectures.
- Domain-specific architectures are based on “accelerators”: Specialized hardware structures designed to efficiently execute dominating computation patterns and improve performance under a given transistor budget.

FSoC Semester Project WS22/23

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FSoC Lab WS22/23 – Hardware Accelerator Design

- Application areas
 - Networking - Security, monitoring, function virtualization
 - Streaming analytics
 - Financial technology (Fintech) - Banking, investment, risk management
 - Genomic sequencing
 - Video transcoding, media processing
 - Machine learning
 - Anything with significant parallelism
 - ...

FSoC Lab WS22/23 – Hardware Accelerator Design

- This trend is not limited to large-scale computing platforms. Specialized and application specific computing units have become common place in modern embedded computer architectures:
 - The ST Microelectronics **STM32G4** series combines a 32-bit Arm® Cortex®-M4 core (with FPU and DSP instructions) running at 170 MHz combined with **mathematical accelerators**: one for trigonometric calculations and the other for filtering functions.
 - The ST Microelectronics **STM32F479XX** series integrates various cryptographic hardware accelerators (HW accelerator for AES 128, 192, 256, Triple DES, HASH (MD5, SHA-1, SHA-2) and HMAC)
 - Texas Instruments **AWR1443** is an automotive single-chip FMCW radar sensor capable of operation in the 76- to 81-GHz band. The device comes with an integrated ARM R4F processor and a **hardware accelerator** for radar data processing ...
 - The Renesas **RX72T** series of 32-bit MCUs are optimized for motor control. They have dedicated hardware accelerators to perform the complex, high-speed computations required for motor control in robots and other industrial applications.
 - ...
- There will be MCUs with integrated AI accelerators in the near future to enable intelligent IOT/AI at the edge devices ...

FSoC Lab WS22/23 – Cryptographic Hardware Accelerator Design

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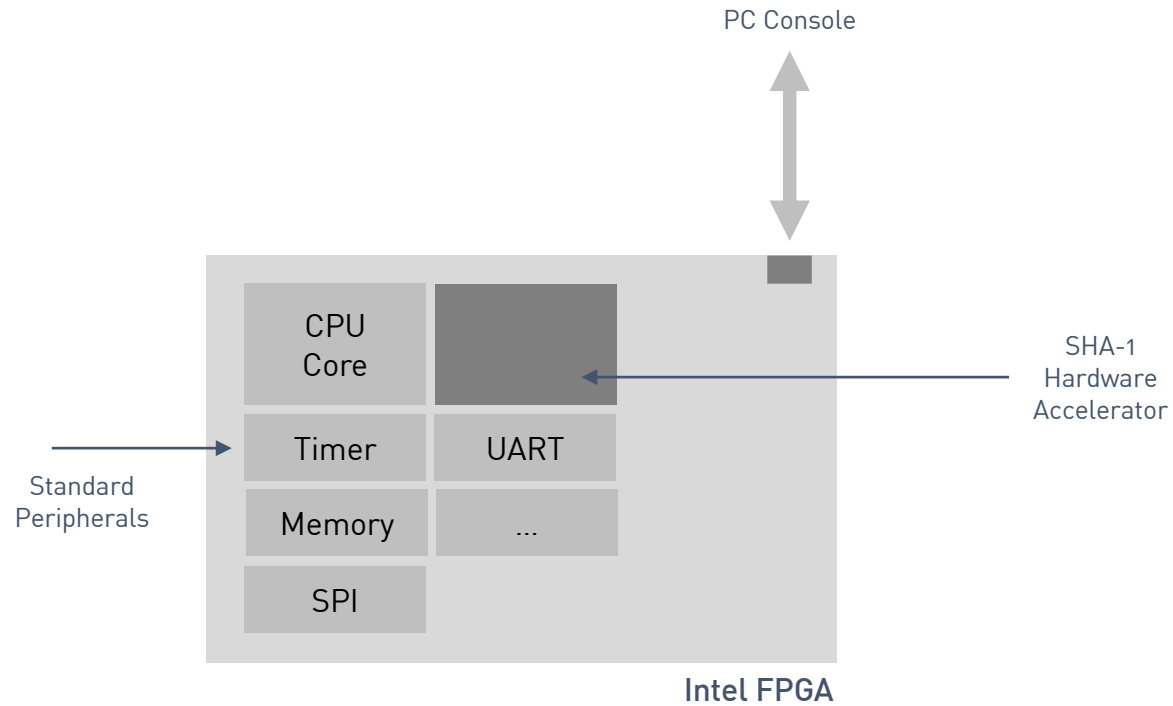
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FSoC Lab WS22/23 – Cryptographic Hardware Accelerator Design

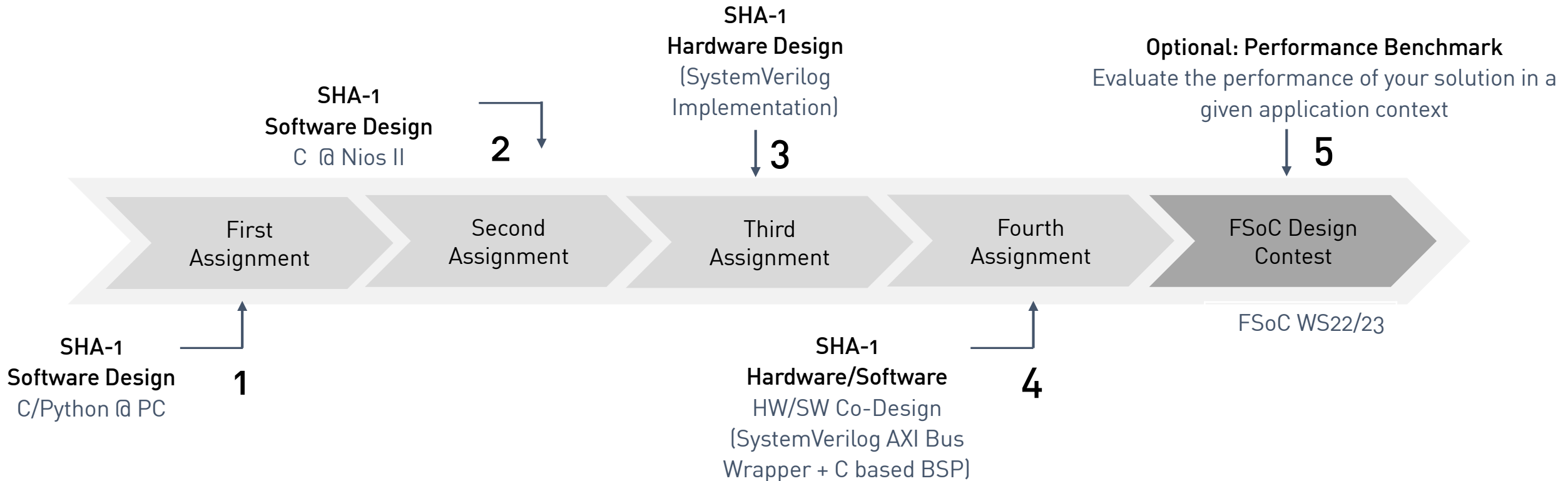
- Our intention is to design an FPGA based Embedded System similar to the ones described before: The task is to analyze, design and implement a cryptographic hash function (SHA-1) in hardware (SystemVerilog) and software (C/C++ bare-metal on the Intel Nios II softcore processor or on the ARM Cortex A9 running Linux).
- The hardware unit finally acts as dedicated accelerator placed in the peripheral-set of the Nios II CPU to accelerate the computation of the SHA-1 algorithm.
- Hash functions are based on simple mathematical recipes. They map data of arbitrary size to a fixed length output (the so called hash).



FSoC Lab WS22/23 – Cryptographic Hardware Accelerator Design



FSoC Lab WS22/23 – Cryptographic Hardware Accelerator Design



FSoC Semester Project WS22/23

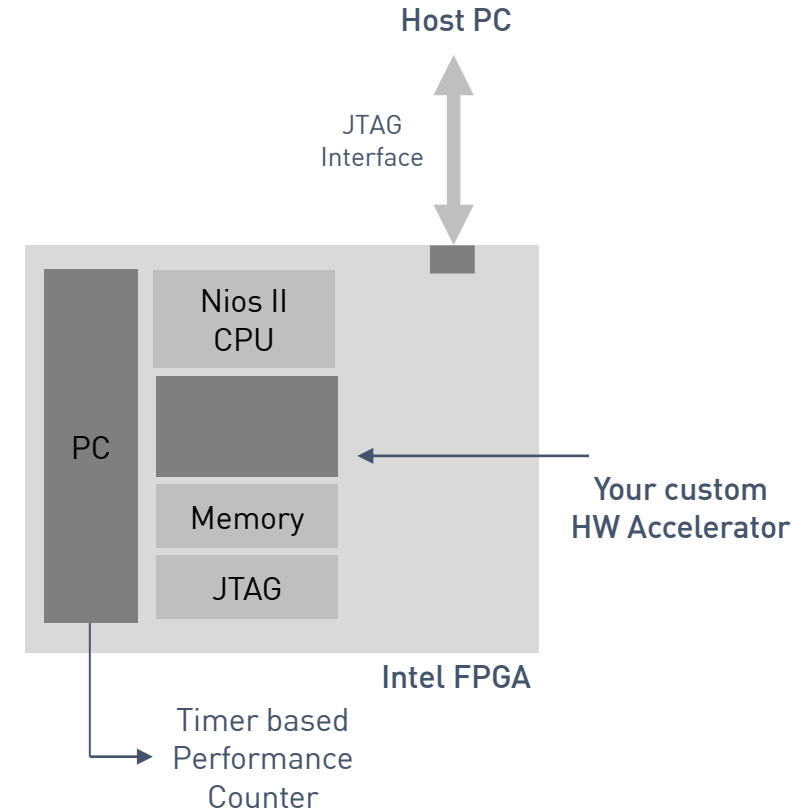
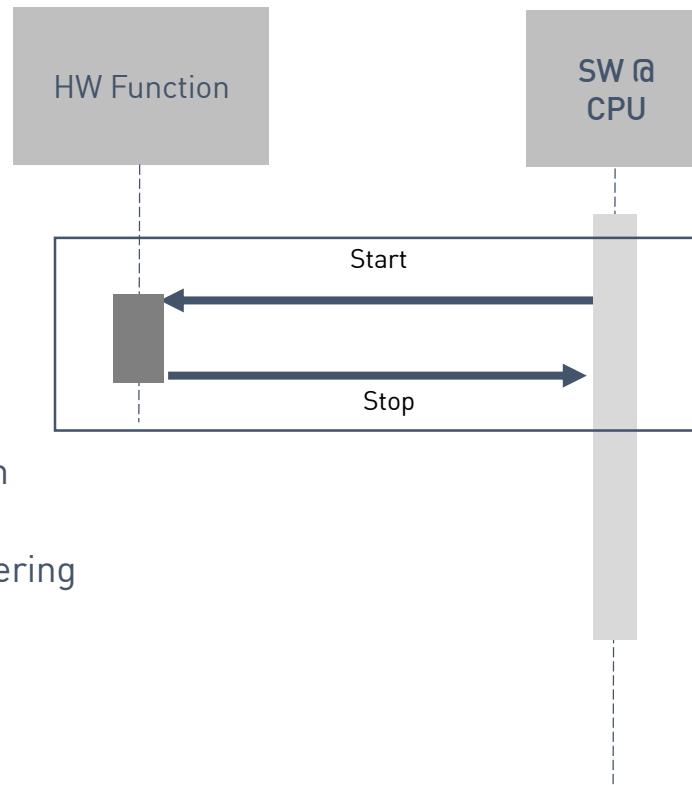
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The FSoC Design Challenge WS22/23

- Use and benchmark your design in a dedicated application context.



- Inspired by Patrick R. Schaumont annual design challenge running at the Professor Bradley Department of Electrical and Computer Engineering Virginia Tech.



Exam and Evaluation

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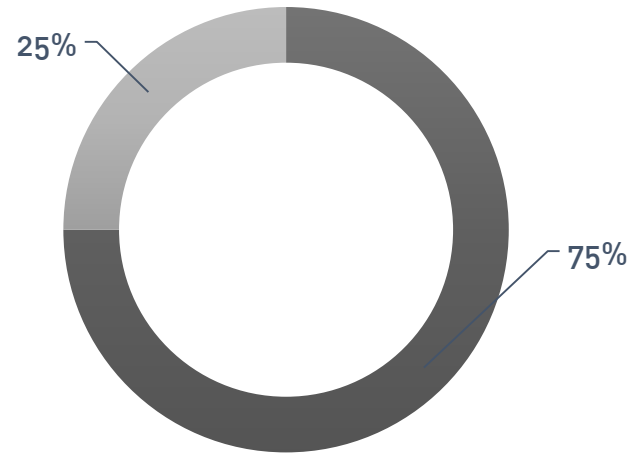
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FSoC Examination WS22/23

- WS22/23 semester grades will be based on the following weights:

Final Grade

- Exam (75%)
- Lab part (25%)



- There is the possibility to get extra credit for the completion of homework assignments.

Recommended Readings

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Recommended Readings

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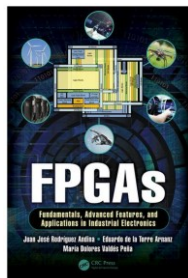
Course related Literature

- Kilts, Steven, “Advanced FPGA Design – Architecture, Implementation, and Optimization”, Wiley-IEEE Press, 2007



- Addressing advanced issues of FPGA (Field-Programmable Gate Array) design and implementation, Advanced FPGA Design: Architecture, Implementation, and Optimization

- Andina, J., Arnanz, E., Valdes, M., “FPGAs: Fundamentals, Advanced Features, and Applications in Industrial Electronics”, CRC Press, 2017



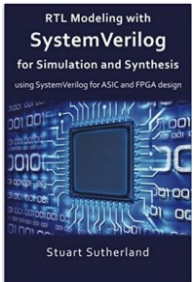
- Explains how and why field-programmable gate arrays (FPGAs) may be used for high-end applications in a variety of domains.
- Analyzes fundamental architectures of FPGA devices, providing case studies and simple design examples
- Presents a state-of-the-art vision of FPGA technologies as well as a mid-term forecast for the future

Recommended Readings

h_da – fbeit - FPGA-based SoC Design

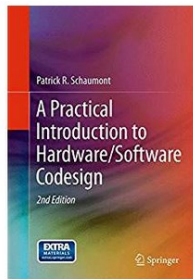
Course related Literature

- Sutherland, S., “RTL Modeling with SystemVerilog for Simulation and Synthesis: Using SystemVerilog for ASIC and FPGA Design”, CreateSpace Independent Publishing Platform, 2017



- This book is both a tutorial and a reference for engineers who use the SystemVerilog Hardware Description Language (HDL) to design ASICs and FPGAs.
- The book shows how to write SystemVerilog models at the Register Transfer Level (RTL) that simulate and synthesize correctly, with a focus on proper coding styles and best practices.

- Schaumont, P. R., “A Practical Introduction to Hardware/Software Codesign”, Springer, 2nd Edition, 2014



- This textbook serves as an introduction to the subject of embedded systems design, with emphasis on integration of custom hardware components with software.
- The key problem addressed in the book is the following: how can an embedded systems designer strike a balance between flexibility and efficiency?

Recommended Readings

h_da – fbeit - FPGA-based SoC Design

Online Resources

Intel FPGA Technical Training

- <https://community.intel.com/>

All about tooling, FPGAs and HDLs

Intel FPGA University Program

- <https://software.intel.com/content/www/us/en/develop/topics/fpga-academic.html>

Tutorials for Digital Logic Design, Computer Organization and Embedded Systems

RocketBoards

- <https://rocketboards.org/>

Community home for open source collaboration aimed at Linux development for SoC FPGAs

FSoC WS22/23 in a Nutshell ...

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Introduction to FPGA based System on Chip Design

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FSoC WS22/23 in a Nutshell ...

- Everything starts with ...



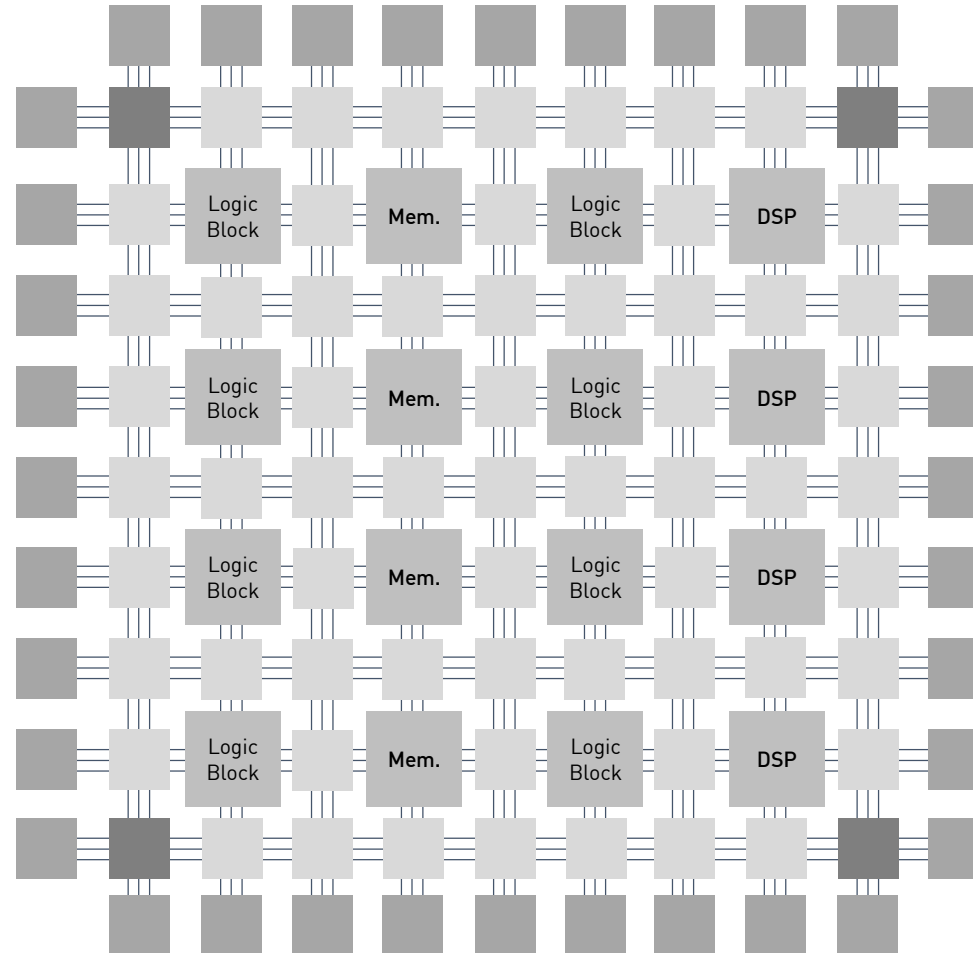
Intel FPGA

Introduction to FPGA based System on Chip Design

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FSoC WS22/23 in a Nutshell ...

- Simplified FPGA (logic-centric) Structure

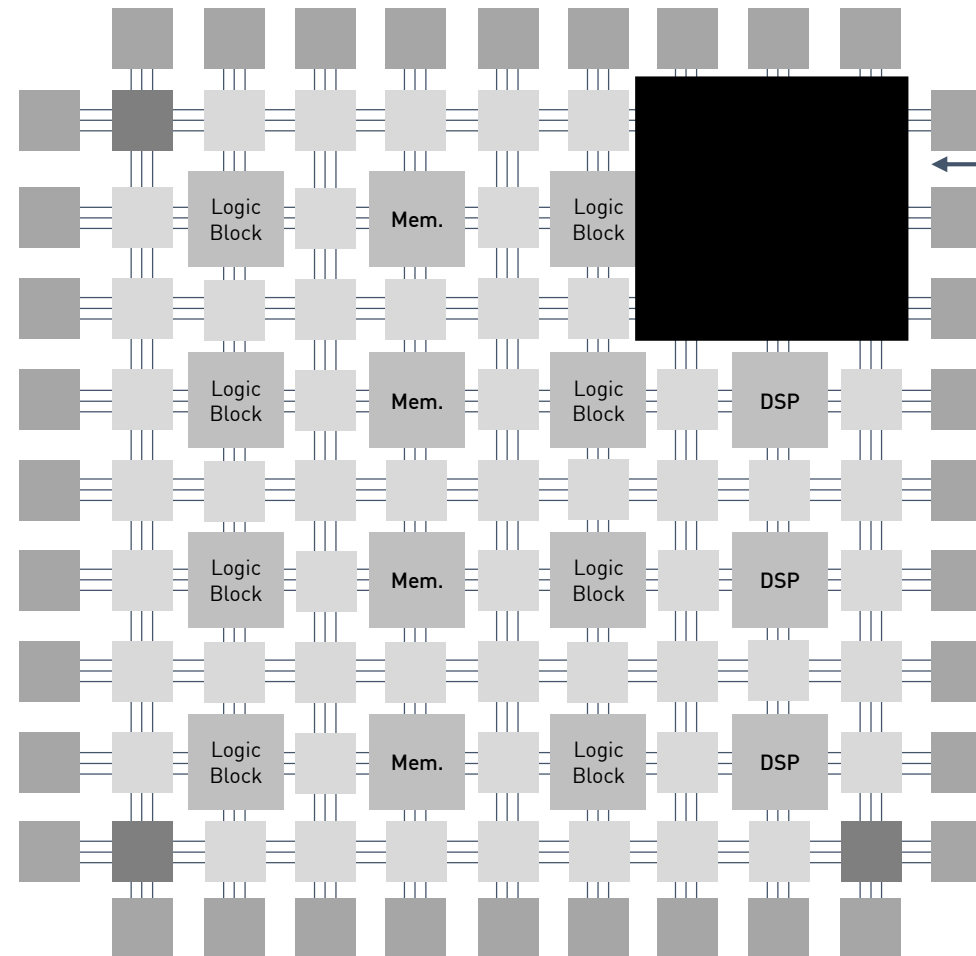


Introduction to FPGA based System on Chip Design

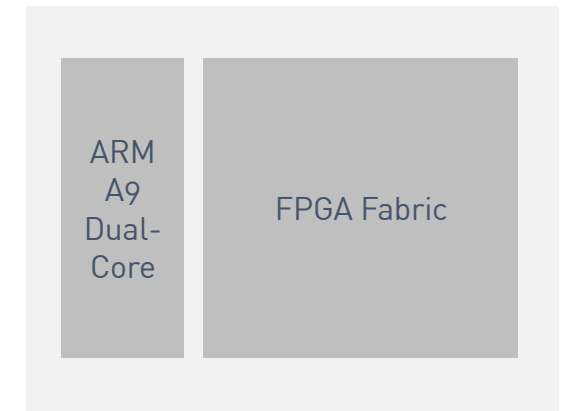
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FSoC WS22/23 in a Nutshell ...

- Simplified FPGA (SoC-centric) Structure



Hard Processor
Systems



Intel SoC FPGA

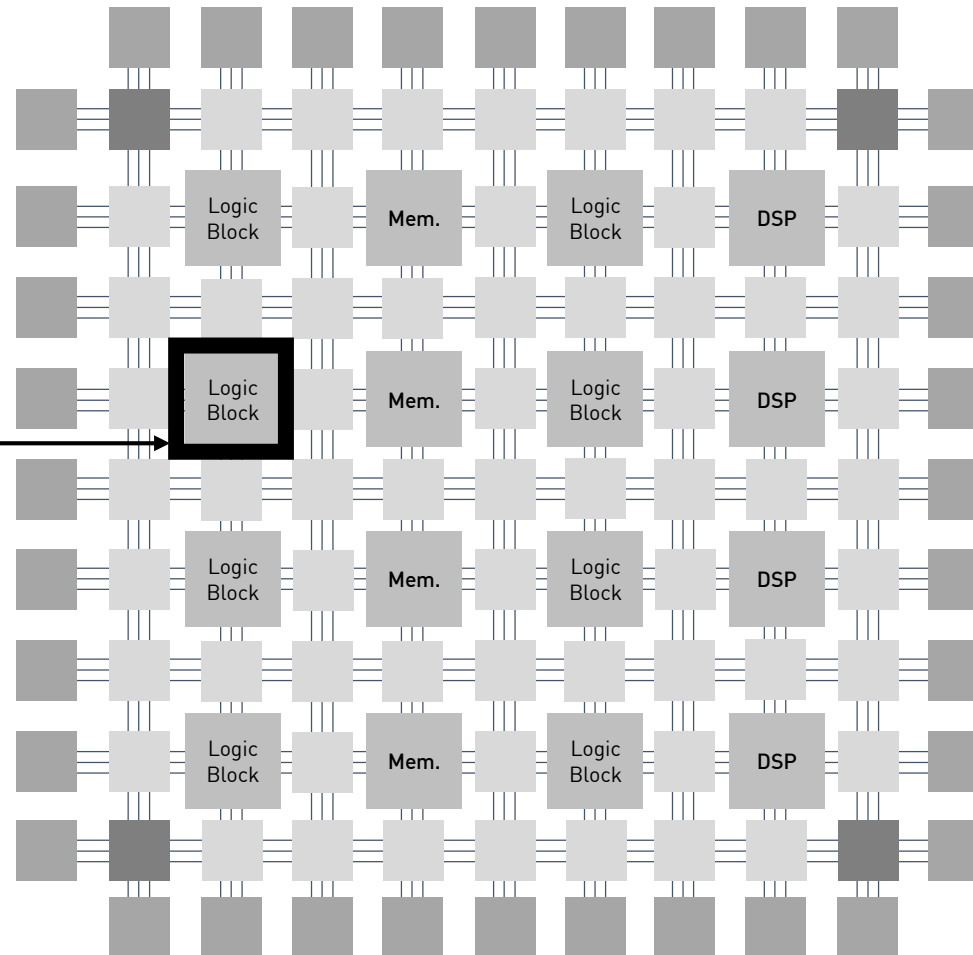
Introduction to FPGA based System on Chip Design

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FSoC WS22/23 in a Nutshell ...

- Simplified FPGA (logic-centric) Structure

Logic Blocks

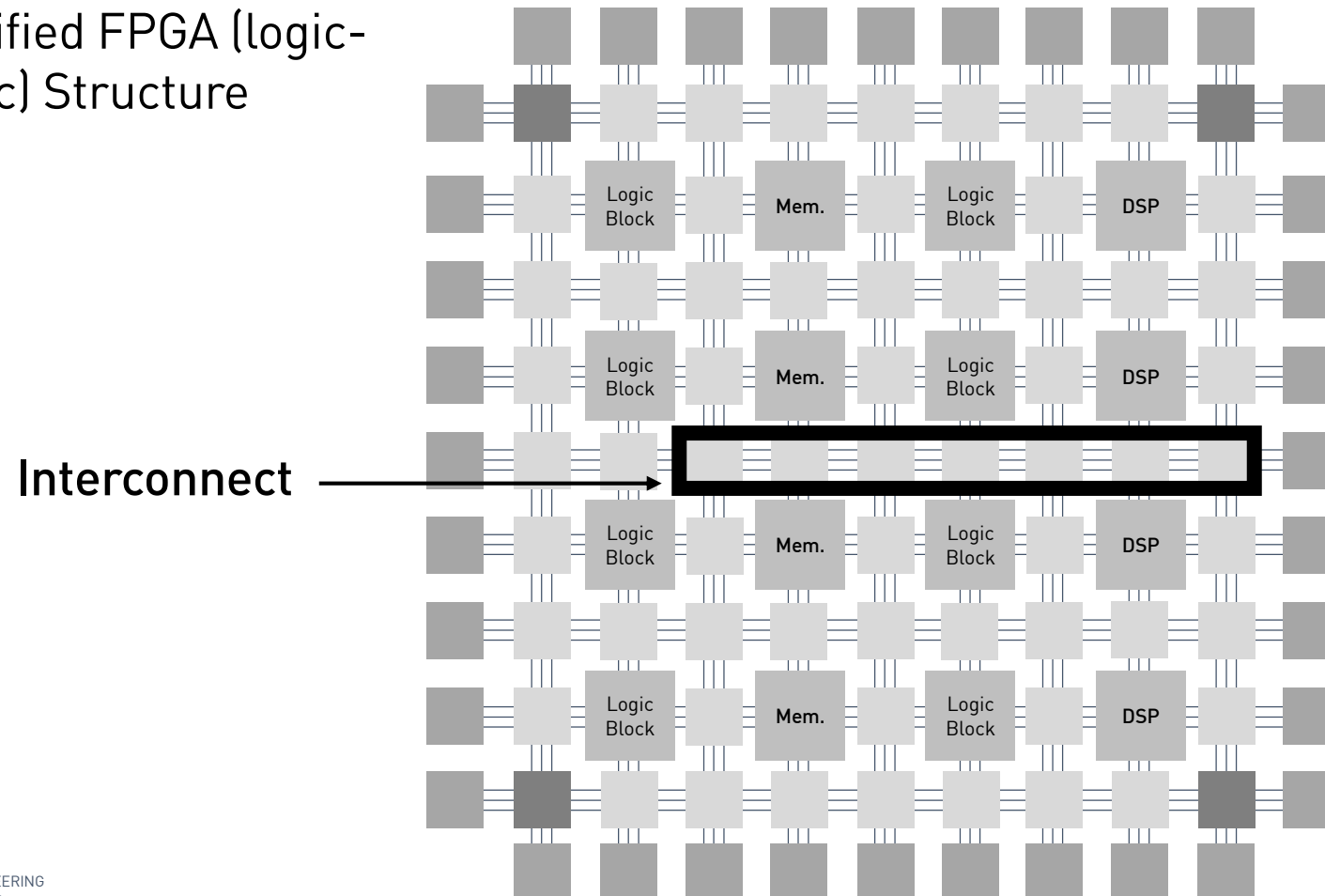


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FSoC WS22/23 in a Nutshell ...

- Simplified FPGA (logic-centric) Structure

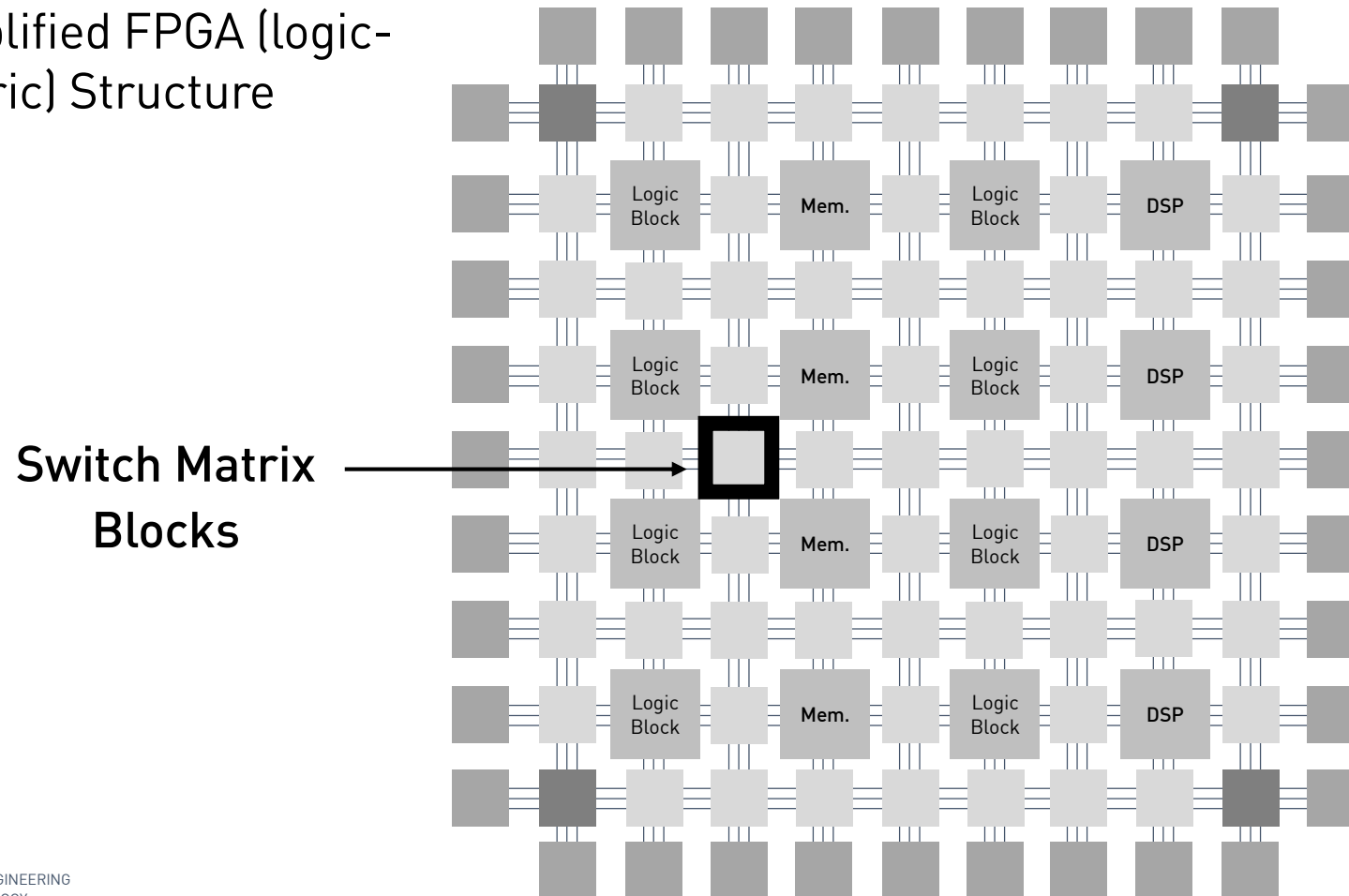


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FSoC WS22/23 in a Nutshell ...

- Simplified FPGA (logic-centric) Structure



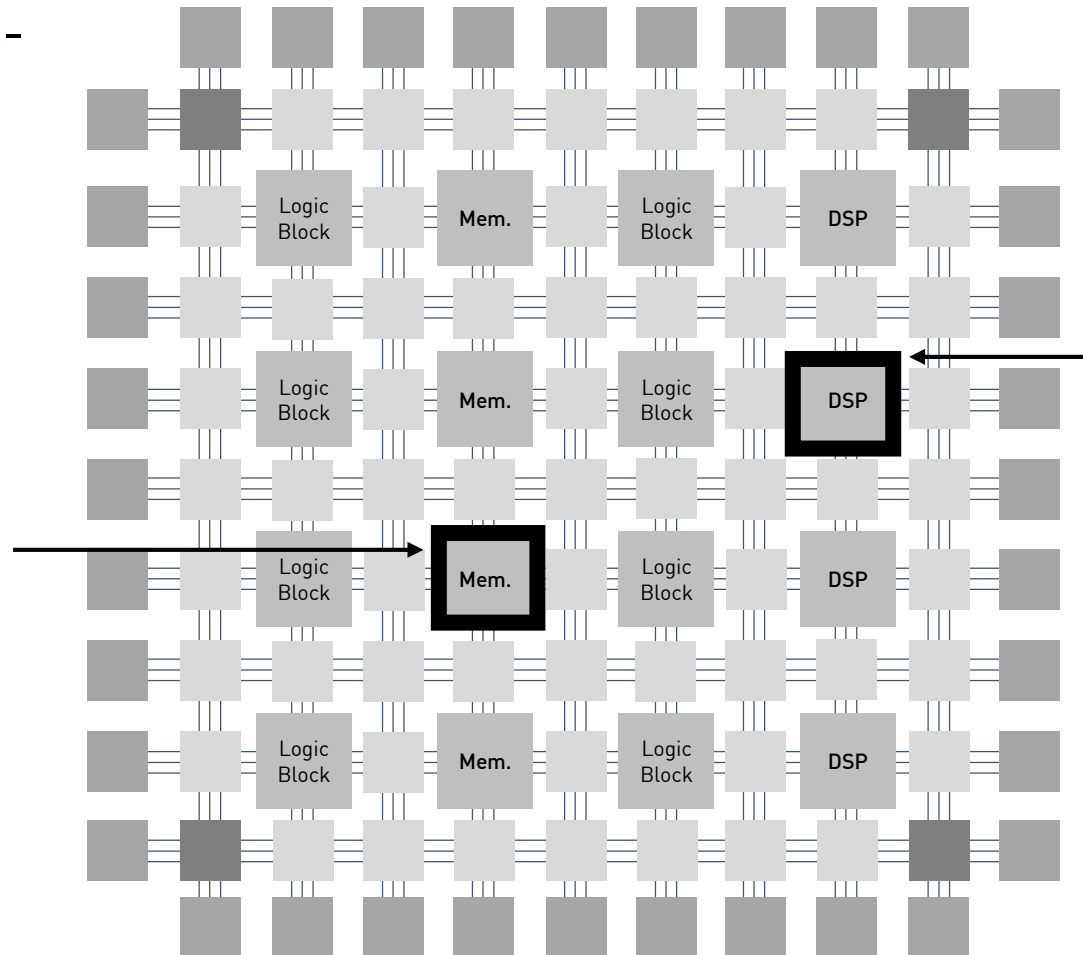
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FSoC WS22/23 in a Nutshell ...

- Simplified FPGA (logic-centric) Structure

Dedicated
Functionality
(Memory)



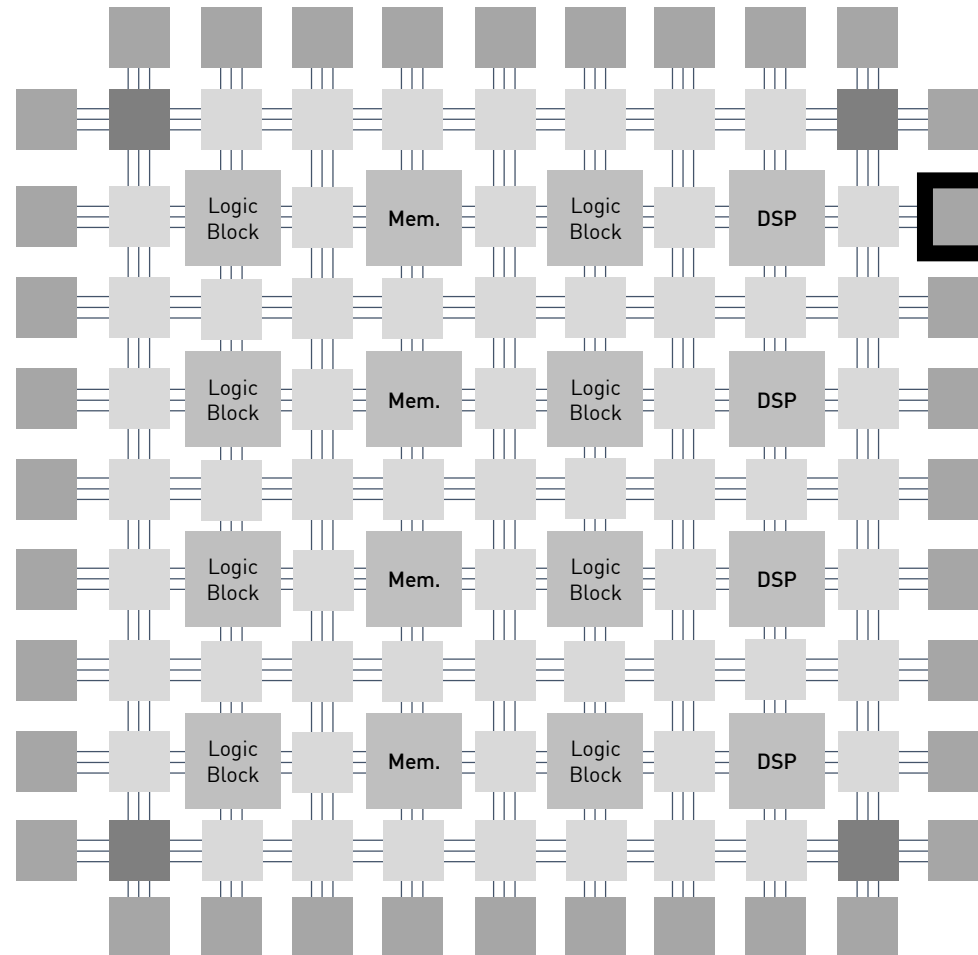
Dedicated
Functionality
(Multiplier)

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FSoC WS22/23 in a Nutshell ...

- Simplified FPGA (logic-centric) Structure



I/O Blocks

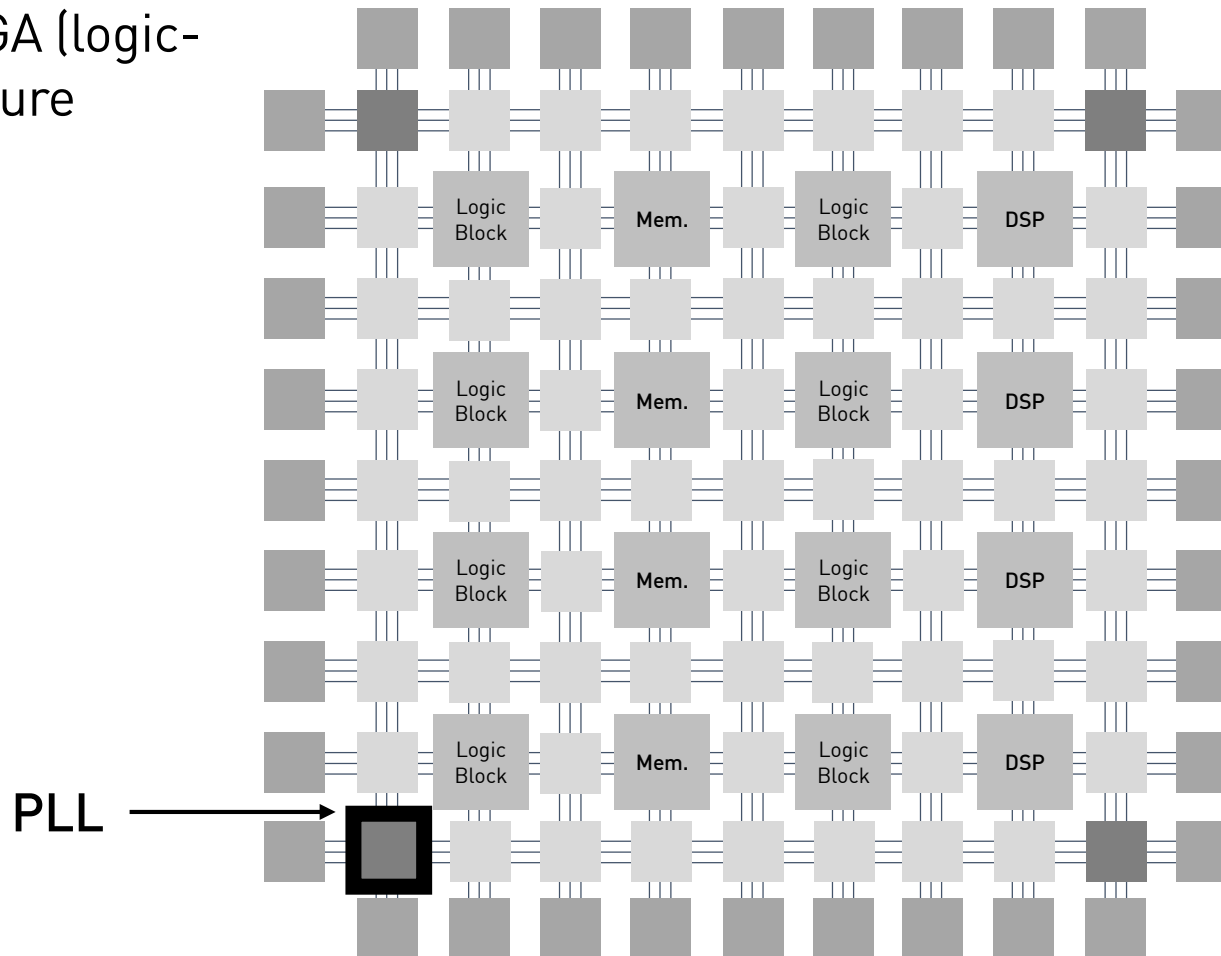
- Supporting various I/O standards

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FSoC WS22/23 in a Nutshell ...

- Simplified FPGA (logic-centric) Structure



FSoC WS22/23 – Mapping HDL to FPGA logic cells

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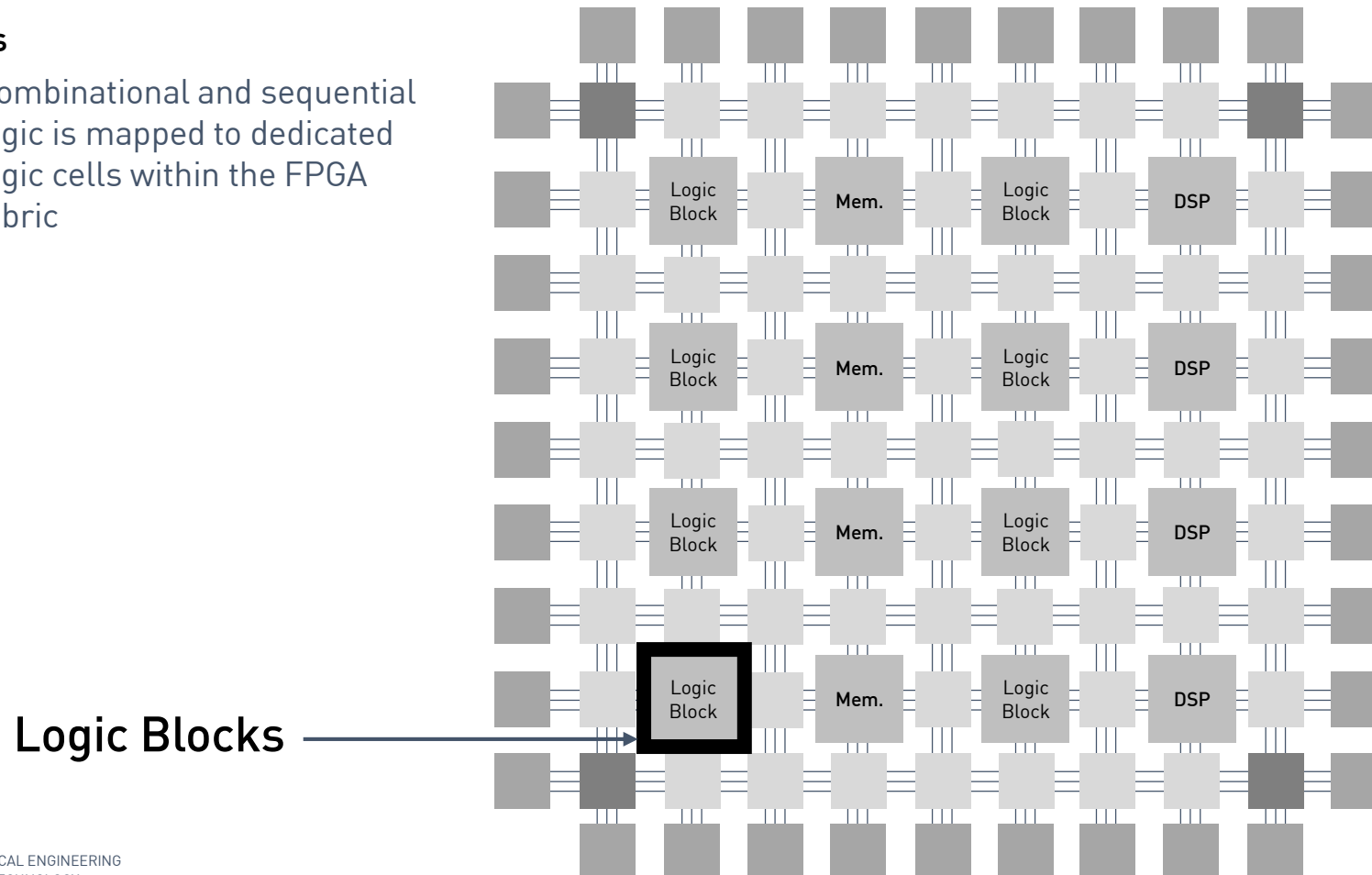
Introduction to FPGA based System on Chip Design

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Basic FPGA Logic Cell Structure

Notes

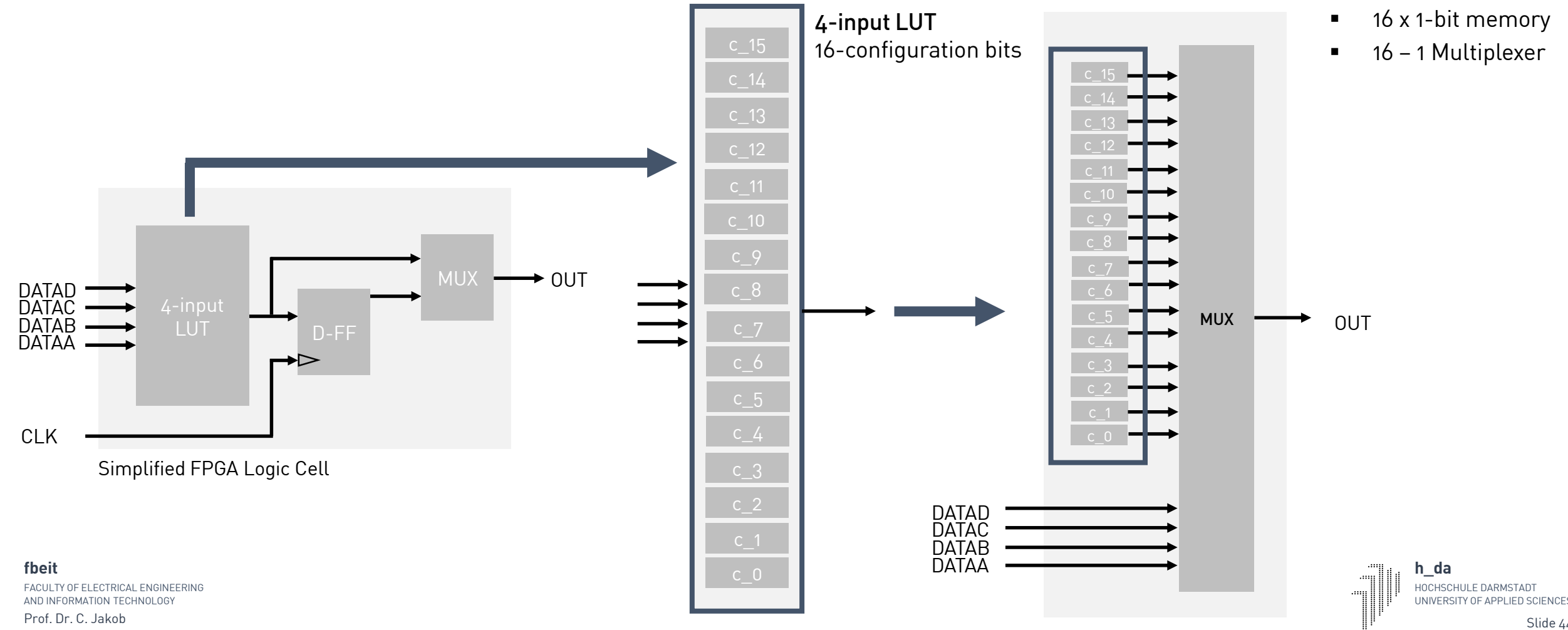
- Combinational and sequential logic is mapped to dedicated logic cells within the FPGA fabric



Introduction to FPGA based System on Chip Design

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Basic FPGA Logic Cell Structure



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4-input LUT

- 16 x 1-bit memory
- 16 – 1 Multiplexer

Diagram components and connections:

- 4-input LUT:** A vertical block with 16 inputs labeled c_{15} through c_0 .
- MUX (Multiplexer):** A large vertical block receiving inputs from the LUT and four data inputs: DATAD, DATA C, DATAB, and DATA A.
- D-FF (D-type Flip-Flop):** Receives the output of the first MUX and is clocked by CLK. Its output feeds the second MUX.
- Second MUX:** Receives the output of the D-FF and a single configuration bit c_{16} (labeled "Mux - Single configuration bit"). Its output is labeled OUT.

Mapping simple logic functions to the Intel Cyclone IV/X FPGA

FPGA-based SoC Design

Introduction to FPGA based System on Chip Design

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Greater than or equal to 9 ...

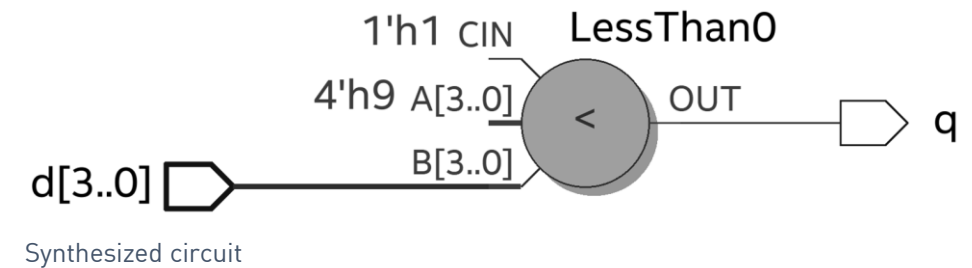
```
1. module bool_function_v1(
2.     input logic [3:0] d,
3.     output logic q
4. );
5. // ### logic function implementation ...
6. assign q = (d >= 9) ? 1'b1 : 1'b0;
7.
8. endmodule
```

1/1

[SystemVerilog] Source Code: bool_function_v1.sv - logic function >= 9

Use case:

- Logic function implementation: Proof if a 4-bit logic input is greater than or equal to 9



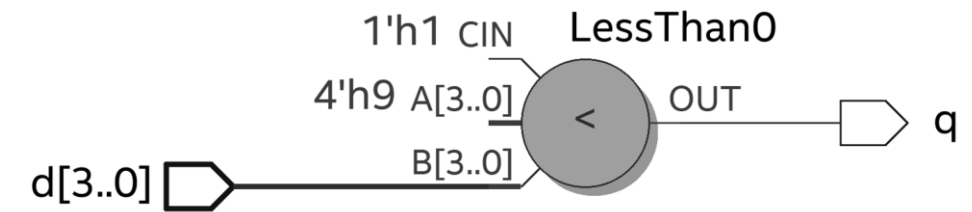
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Greater than or equal to 9 ...

```
1. module bool_function_v2(  
2.     input logic [3:0] d,  
3.     output logic q  
4. );  
5. // ### logic function implementation ...  
6. always_comb  
7.     if (d >= 9)  
8.         q = 1'b1;  
9.     else  
10.        q = 1'b0;  
11.  
12. endmodule
```

1/1



Synthesized circuit

[SystemVerilog] Source Code: bool_function_v2.sv - logic function ≥ 9

Notes

- Leads to the exact same synthesis result ...

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```
1. module bool_function_v2(
2.     input logic [3:0] d,
3.     output logic q
4. );
5. // ### logic function implementation ...
6. always_comb
7.     if(d >= 9)
8.         q = 1'b1;
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10.        q = 1'b0;
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```

Slide 49

Introduction to FPGA based System on Chip Design

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Greater than or equal to 9 ...

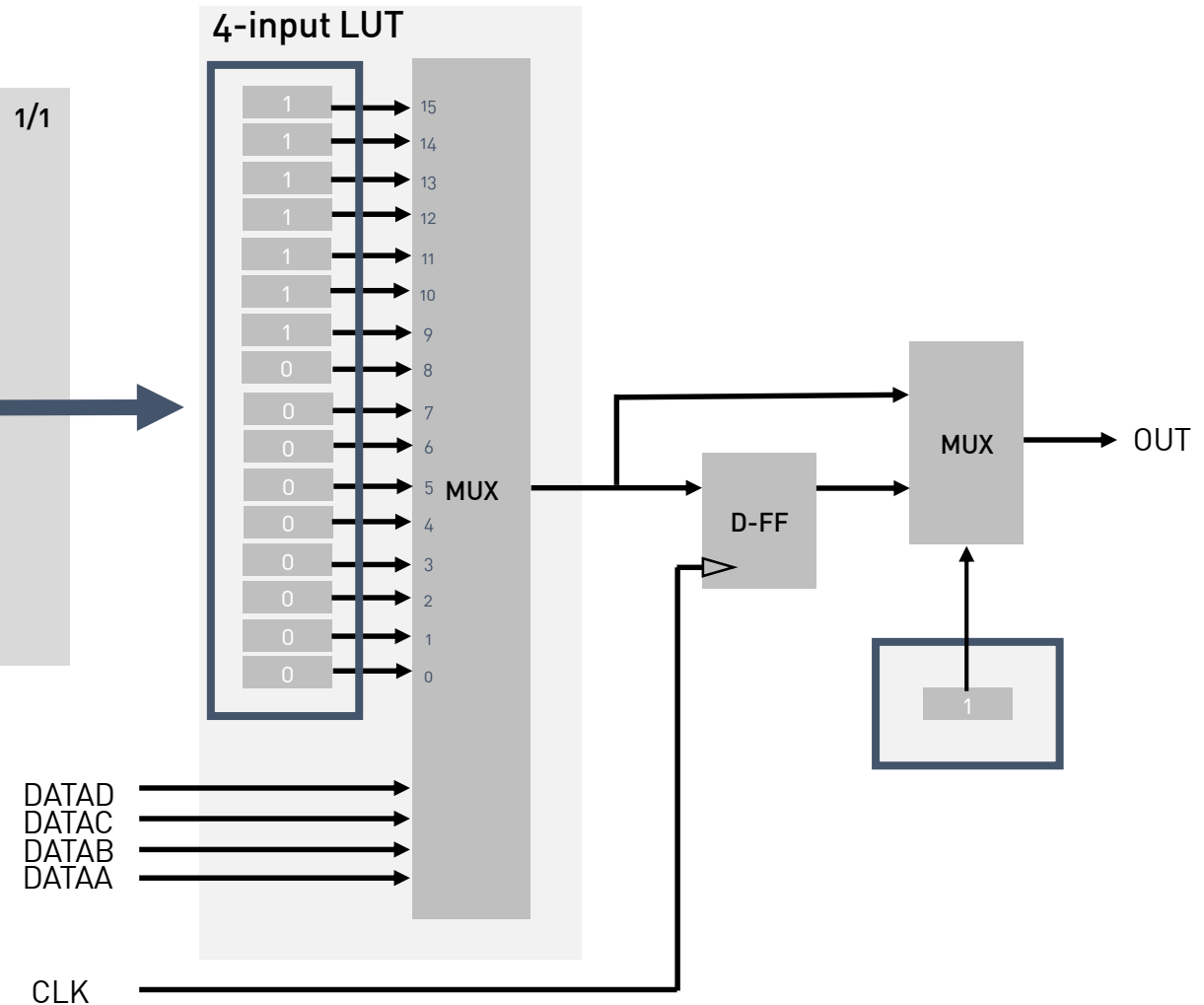
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9.     else  
10.        q = 1'b0;  
11.  
12. endmodule
```

1/1

[SystemVerilog] Source Code: bool_function_v2.sv - logic function ≥ 9

Notes

- Leads to the exact same synthesis result ...



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SoC FPGA based System Design

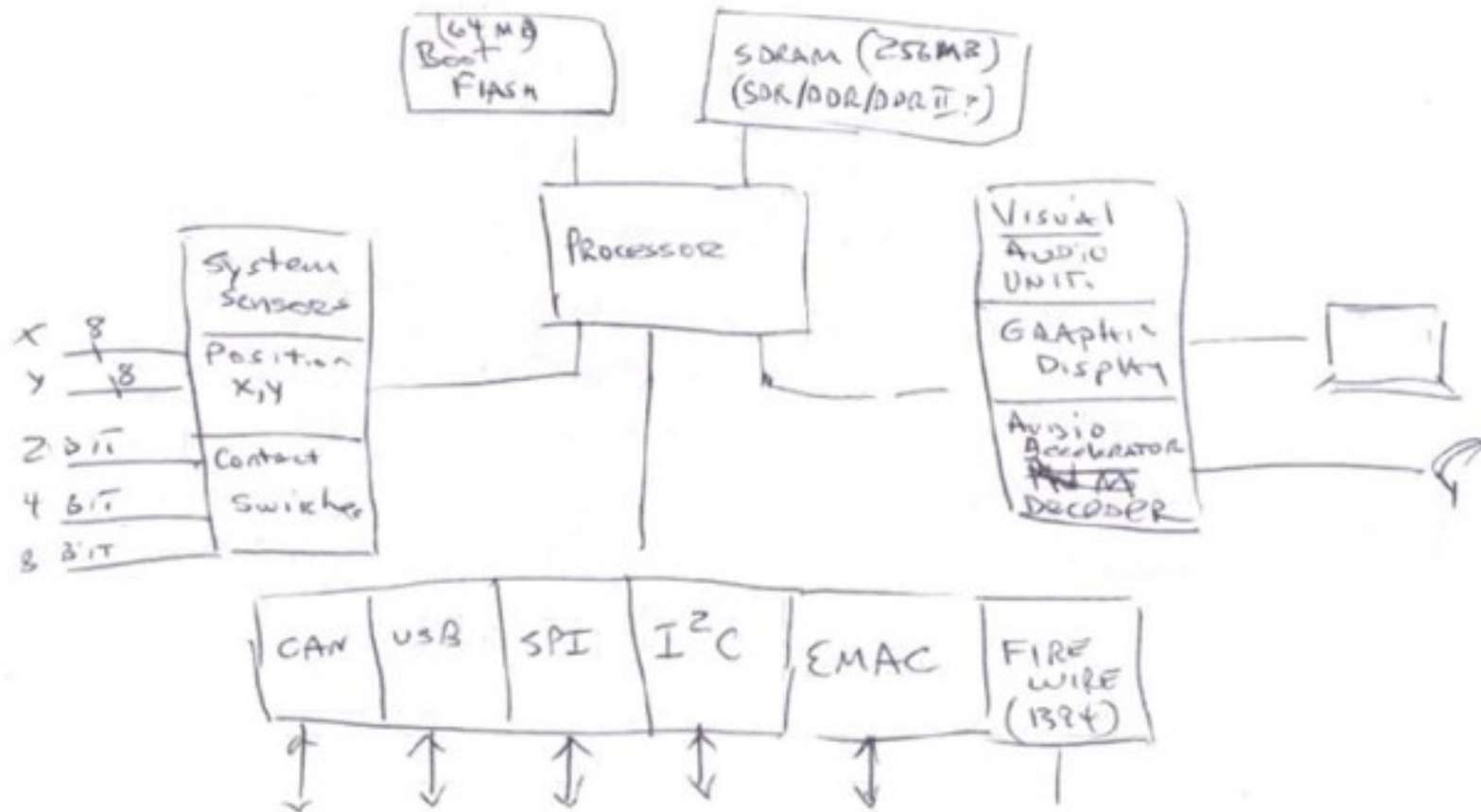
FPGA-based SoC Design

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FSoC WS22/23 in a Nutshell ...

- Everything starts with pencil and paper ...

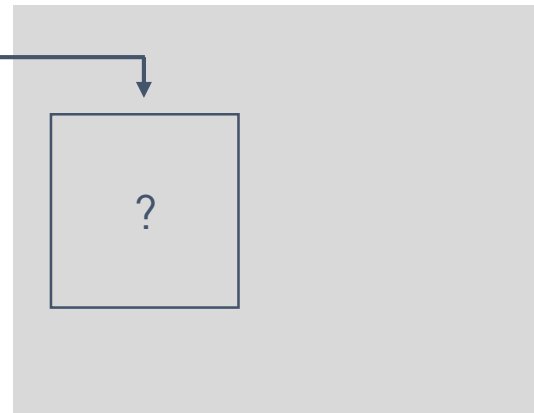


FSoC WS22/23 in a Nutshell ...

- This lecture addresses questions such as ...

First steps

- How to design and describe systems for the implementation on a FPGA-based System platform ...



Intel FPGA

Introduction to FPGA based System on Chip Design

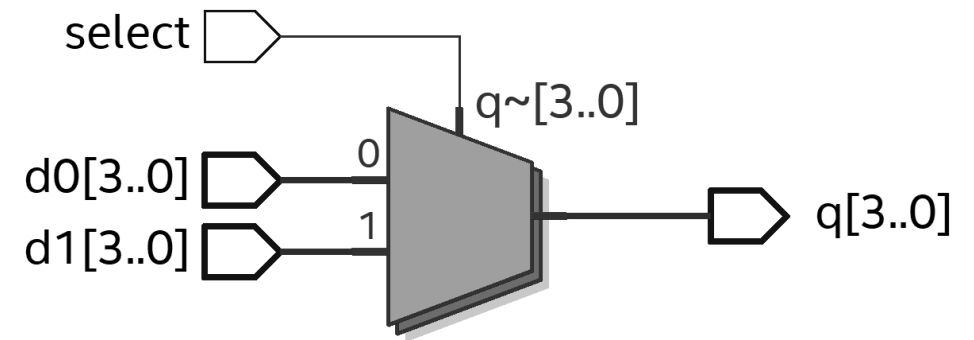
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FSoC WS22/23 in a Nutshell ...

- Everything starts with ...

```
1. module mux_2_4bit(  
2.     input logic [3:0] d0, d1,  
3.     output logic [3:0] q  
4. );  
5.   
6.     always_comb  
7.         if(select == 1'b1)  
8.             q = d1;  
9.         else  
10.            q = d0;  
11. endmodule
```

1/1



[SystemVerilog] Source Code: mux_2_4bit_comb.sv

- How to design and describe systems in SystemVerilog for the implementation on a FPGA-based System platform ...

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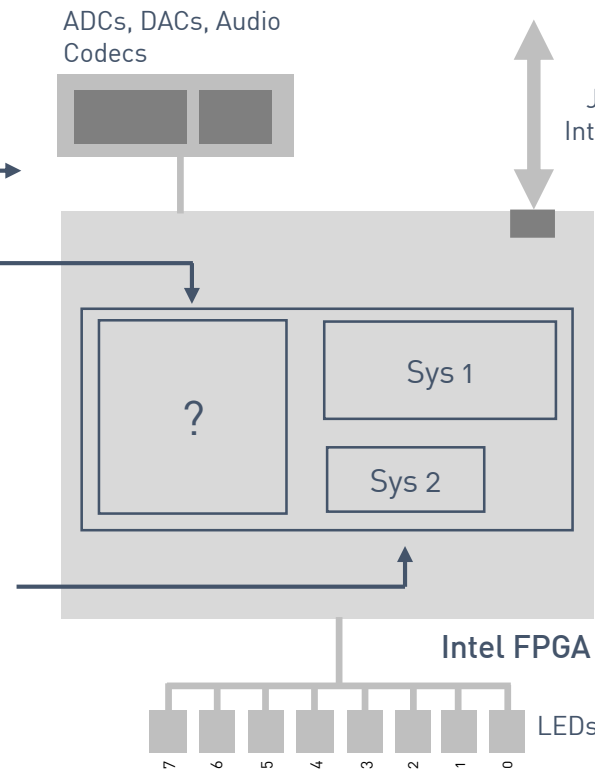
FSoC WS22/23 in a Nutshell ...

- This lecture addresses questions such as ...

- How to interface “more complex” external devices ...

First steps

- How to design and describe systems for the implementation on a FPGA-based System platform ...
- How to create Systems with multiple hardware instances ...



- After successful functional verification, how do we program the FPGA

- How to verify the correct functionality implemented?

- How to interface simple external systems ...

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Softcore CPU based FPGA Embedded Systems

FPGA-based SoC Design

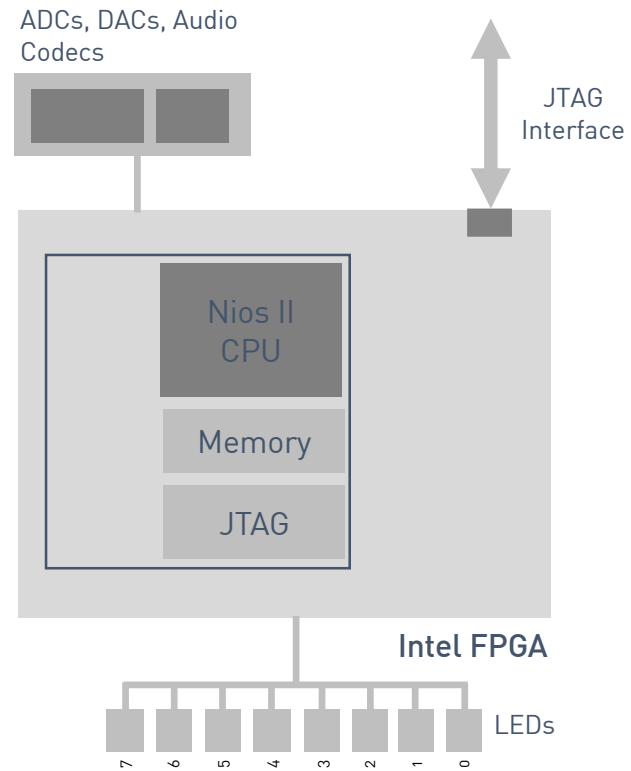
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FSoC WS22/23 in a Nutshell ...

- This lecture addresses questions such as ...

- How to implement a μ C on a FPGA ...

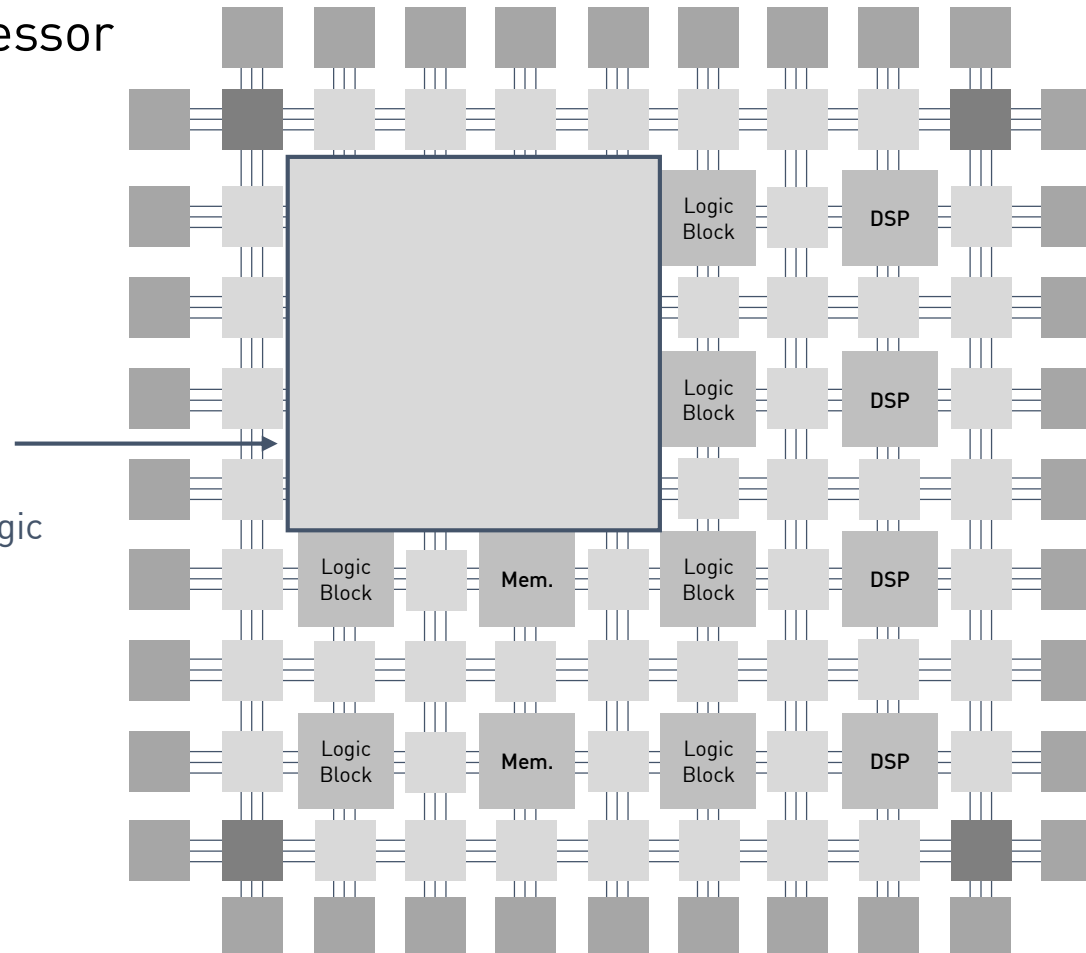


FSoC WS22/23 in a Nutshell ...

- FPGA Softcore Processor

**Softcore
Processor**

Mapped to FPGA Logic
Blocks



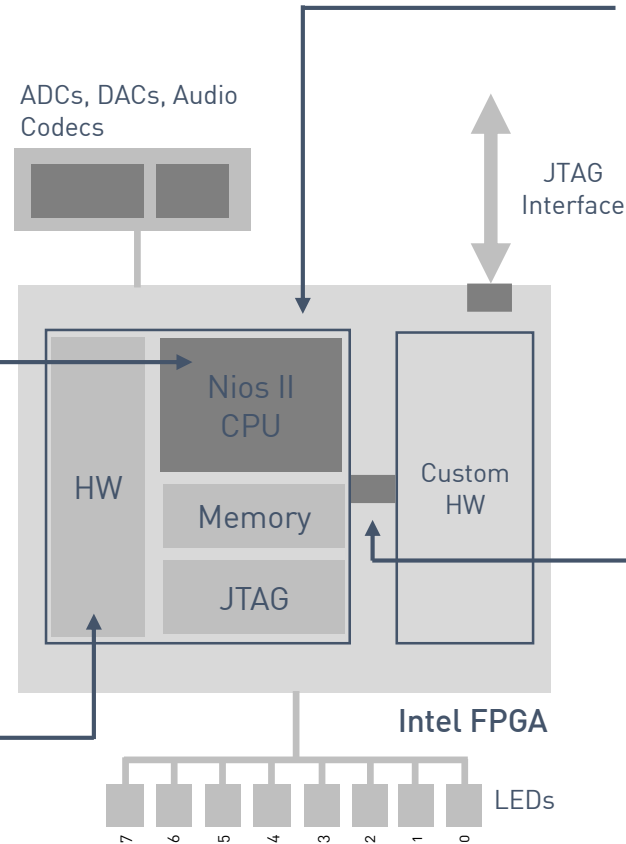
Introduction to FPGA based System on Chip Design

h_da – fbeit - FPGA-based SoC Design

FSoC WS22/23 in a Nutshell ...

- This lecture addresses questions such as ...

- How to program a μ C implemented on a FPGA ...
- How to implement a μ C on a FPGA ...
- How to integrate custom hardware into the peripheral set of the processor ...



- How to write software drivers for accessing custom HW peripherals

- How to interface custom hardware ...

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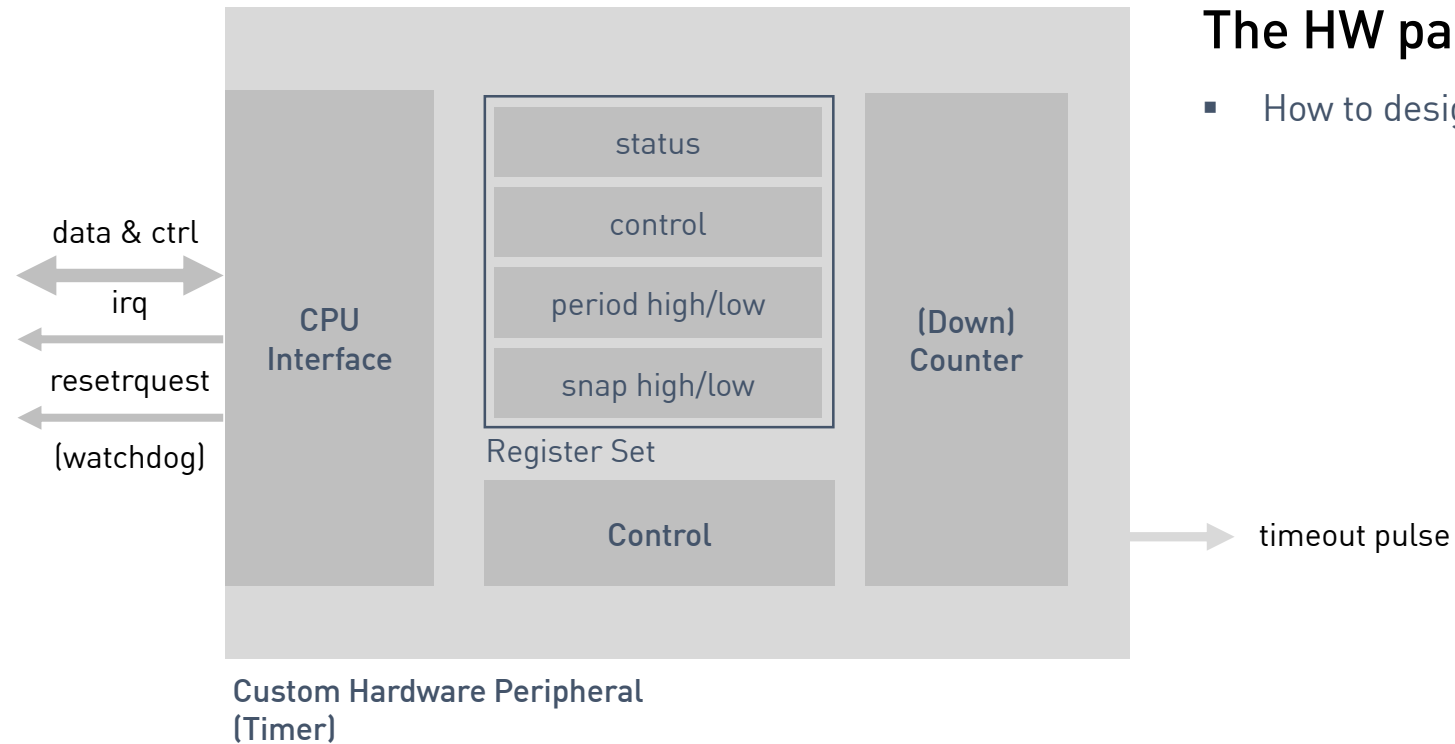
Designing Hardware Peripherals for Softcore CPU Systems

FPGA-based SoC Design

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FSoC WS22/23 in a Nutshell ...



The HW part ...

- How to design custom peripherals ...

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FSoC WS22/23 in a Nutshell ...

```
1. #include "system.h"
2.
3. typedef struct {
4.     __IO unsigned int status;
5.     __IO unsigned int control;
6.     __IO unsigned int periodl;
7.     __IO unsigned int period;
8.     __IO unsigned int snapl;
9.     __IO unsigned int snaph;
10. } timer_t;
11.
12.
13. #define timer_0 (*(volatile timer_t *) (0x8000000 | 0xA0001200 ));
14.
15. ...
16.
17. timer_0.periodl = 0xFFFF;
```

1/1

The SW part ...

- How to access them from SW ...

[C] Source Code: Nios II hardware timer

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FPGA Design - Timing Analysis and Closure

FPGA-based SoC Design

FSoC WS22/23 in a Nutshell ...

- This lecture also addresses questions such as ...

... how to verify the timing of a designed system

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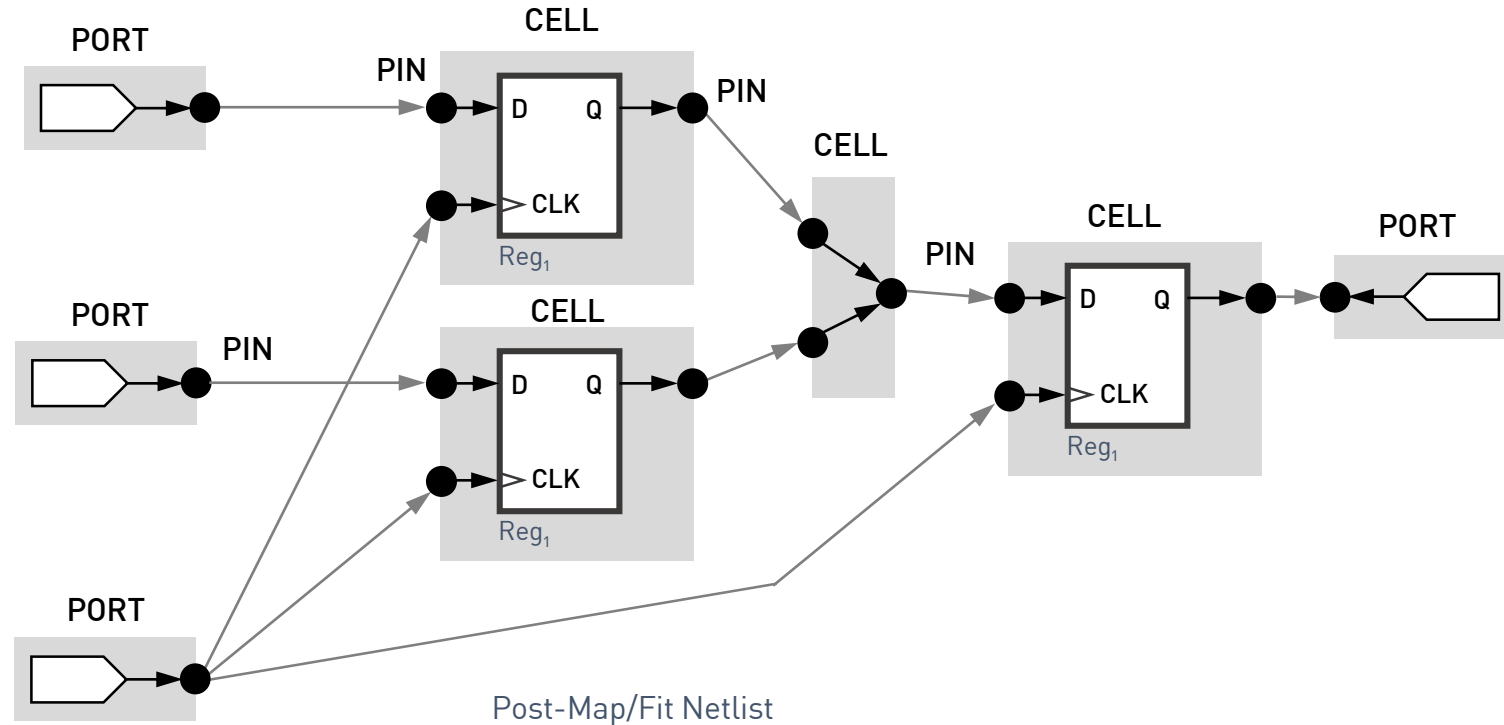
FSoC WS22/23 in a Nutshell ...

■ How to verify timing ...

```
1. module sys_proc(  
2.     input logic [3:0] d0, d1,  
3.     ...  
4. );  
5.   
6.     always_ff@(posedge clk)  
7.         if(reset_n == 1'b1)  
8.             q = 1'b0;  
9.     ...  
10.   
11. endmodule
```

[SystemVerilog] Source Code Excerpt: sys_proc.sv

- The synthesis tool translates the initial HDL design into a netlist which consists of basic components such as cells (logic and/or DFFs), (in and output) ports and pins.

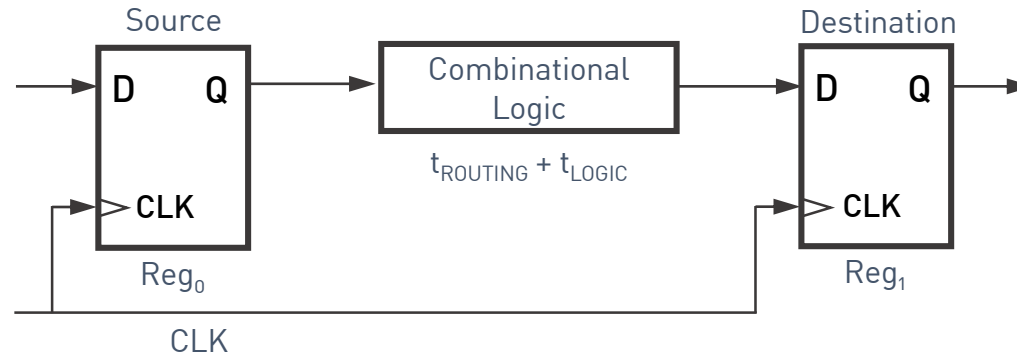


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- Starting point



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Scenario #1 – Timing is met ...

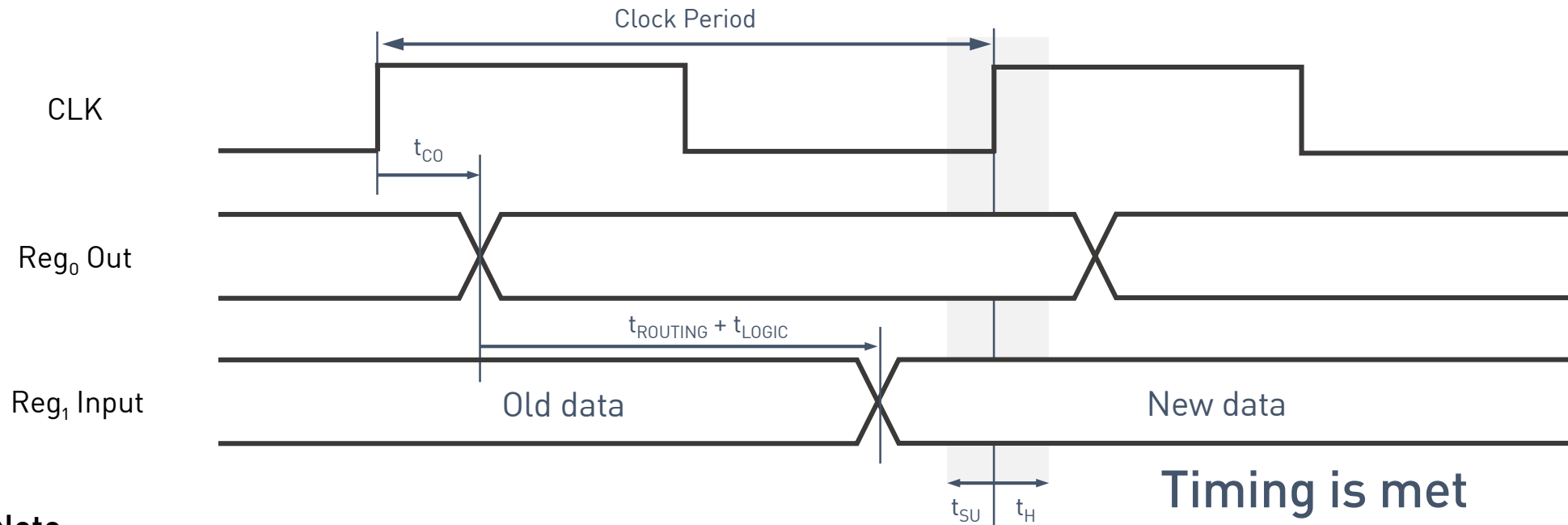
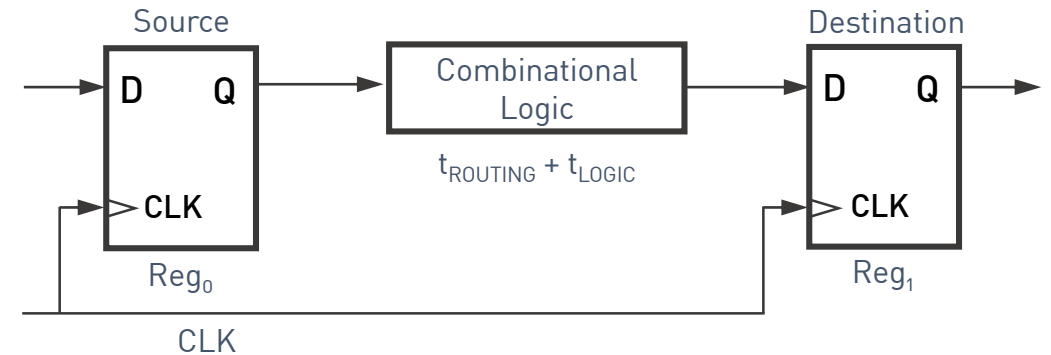
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Scenario: Register transfer takes place within one clock cycle



Note

- Short path delay: $T_{CLK} > t_{CO} + t_{LOGIC} + t_{ROUTING} + t_{SU}$

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Scenario #2 – Timing is not met ...

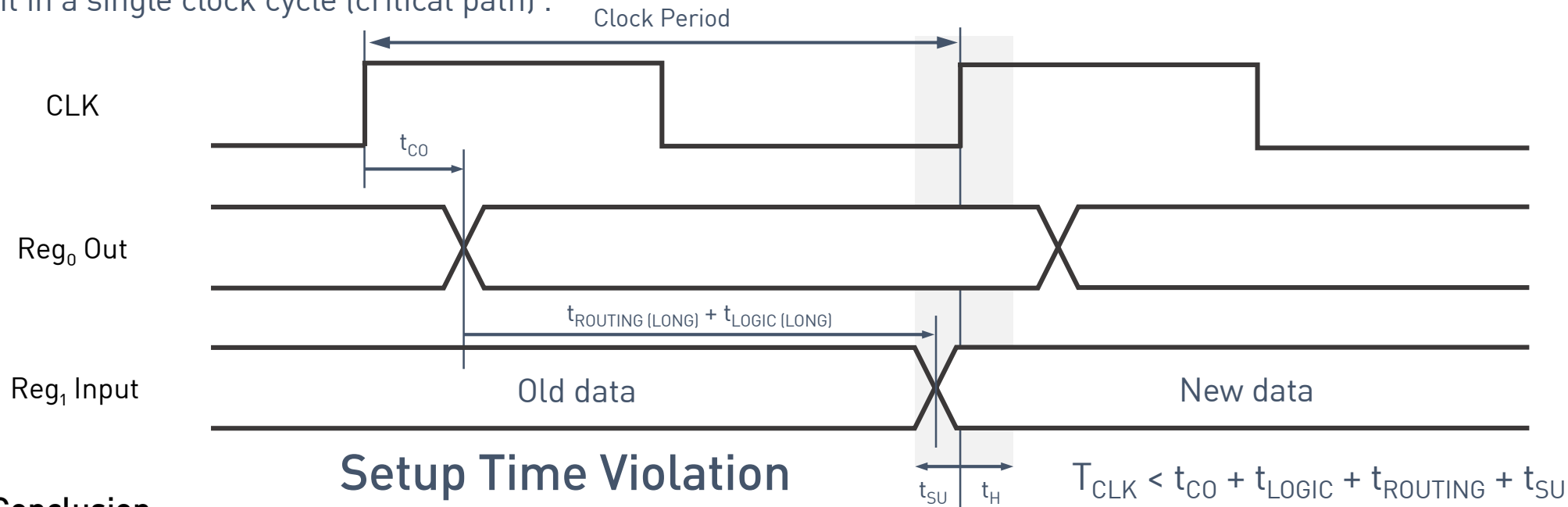
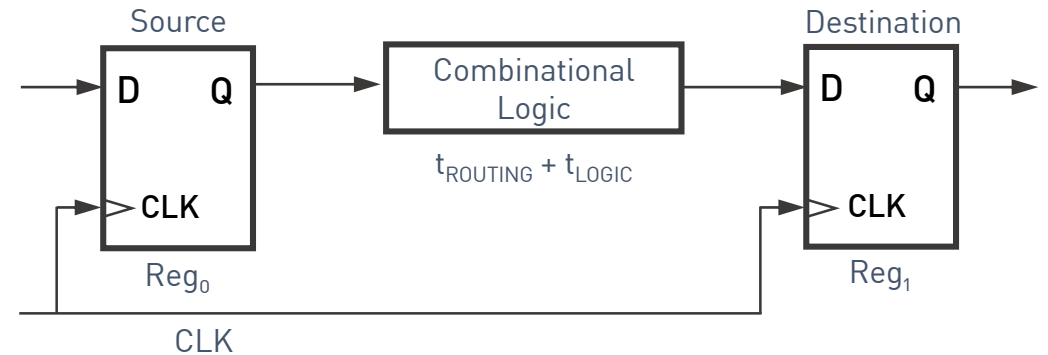
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Scenario: We are considering the longest register to register delay in a design. In the present case, we assume that the register transfer does fit in a single clock cycle (critical path) .



Conclusion

- In order to meet timing, the clock period must be short enough so that the data can reach the destination register before the next rising edge of the clock.

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FPGA Design – Resource and Power Optimization

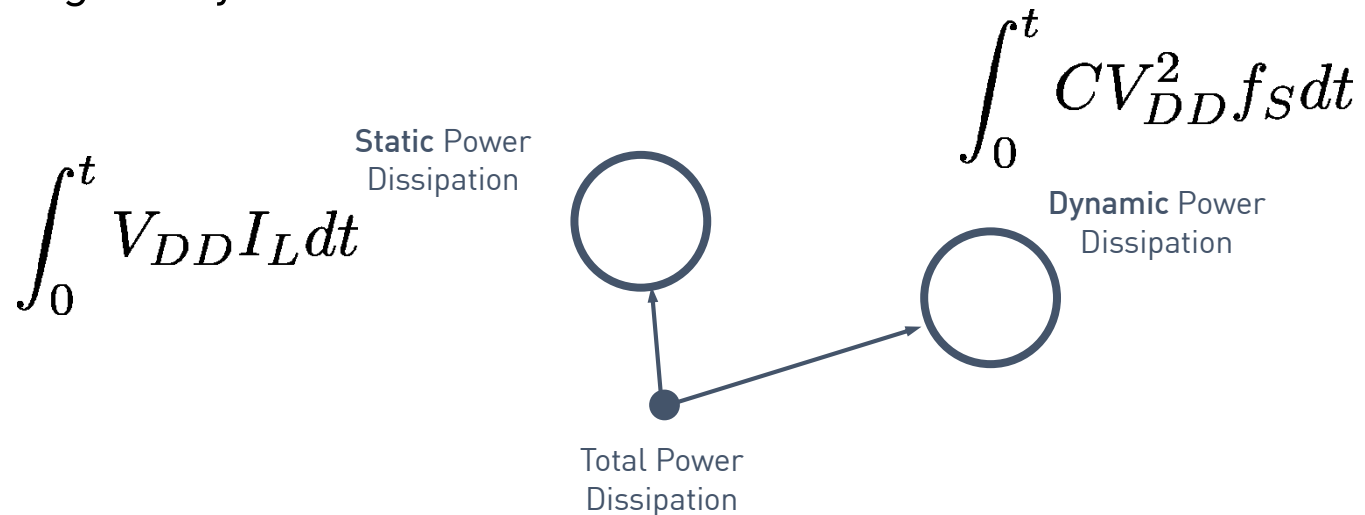
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FSoC WS22/23 in a Nutshell ...

- This lecture also addresses questions such as ...

... how to verify the timing of a designed system

... how to analyze and optimize the power consumption of a designed system



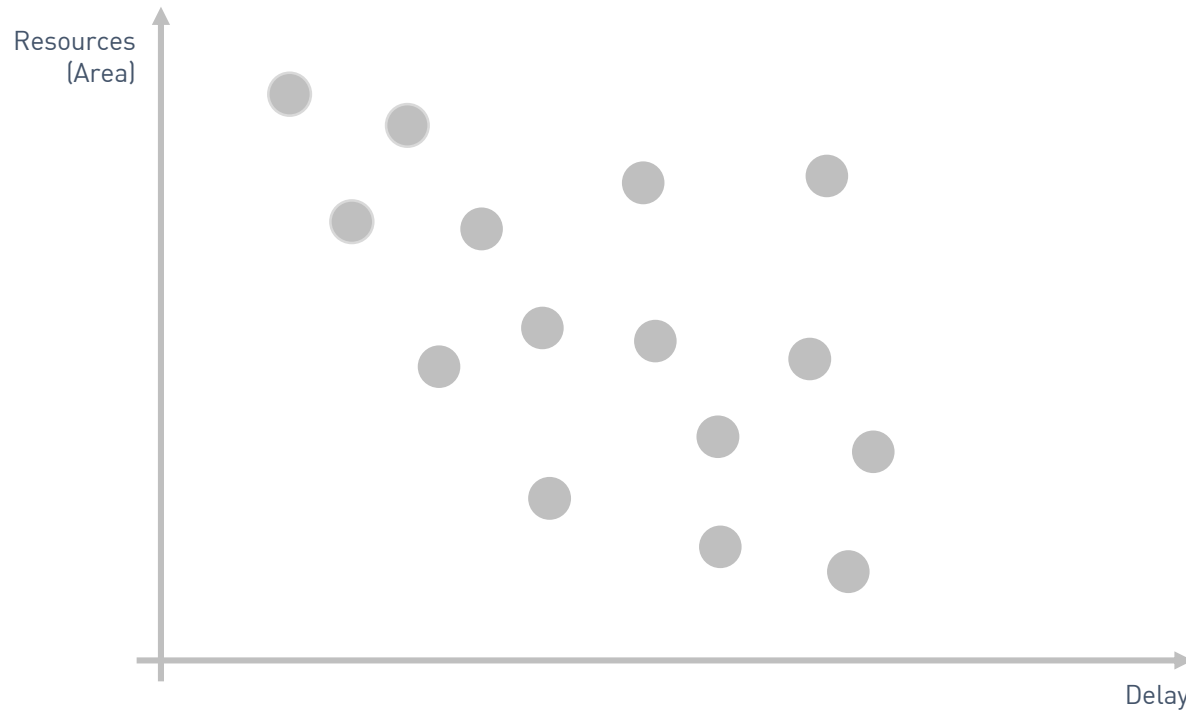
FSoC WS22/23 in a Nutshell ...

- This lecture also addresses questions such as ...
 - ... how to verify the timing of a designed system
 - ... how to analyze and optimize the power consumption of a designed system
 - ... how to efficiently use the design tools in order to achieve **timing**, **resource** or **power** related system constraints

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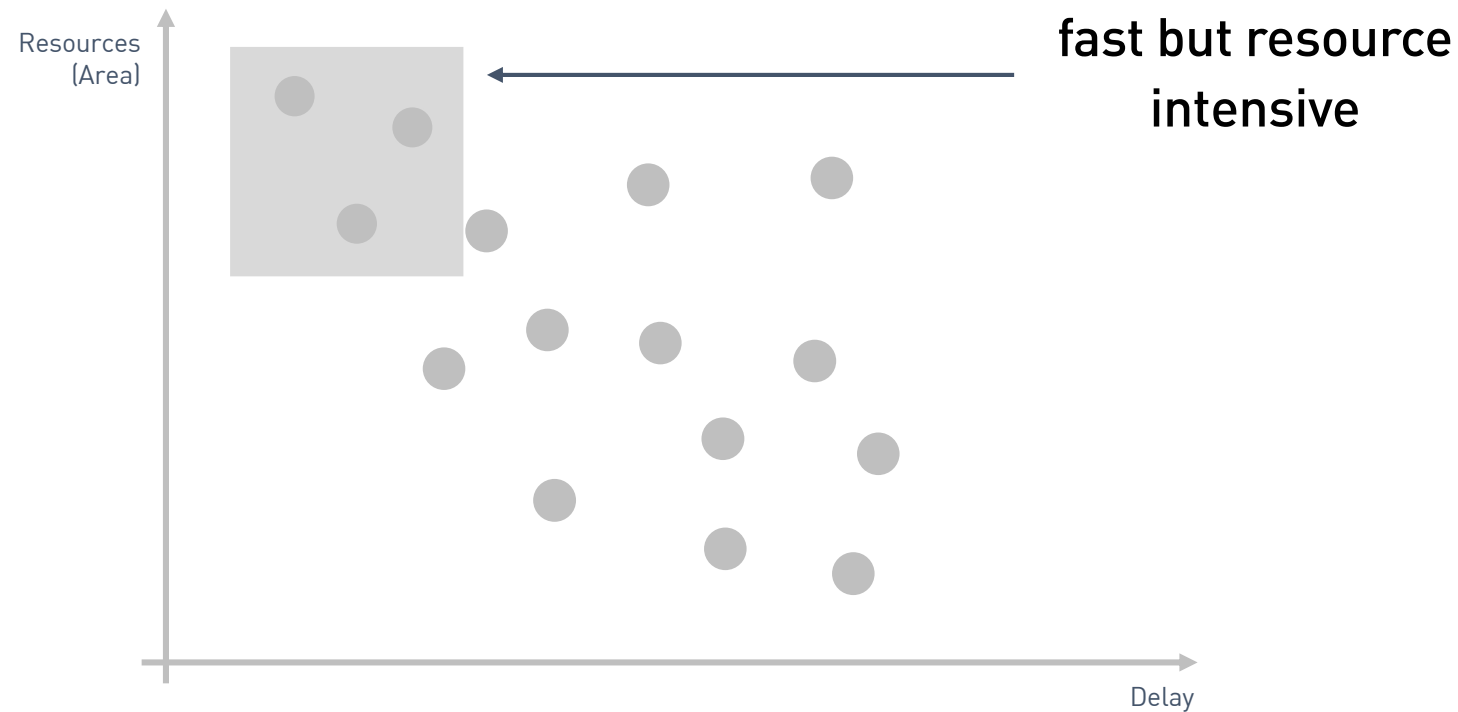


- 15 different designs, each implementing the same functionality
- Which one to choose?

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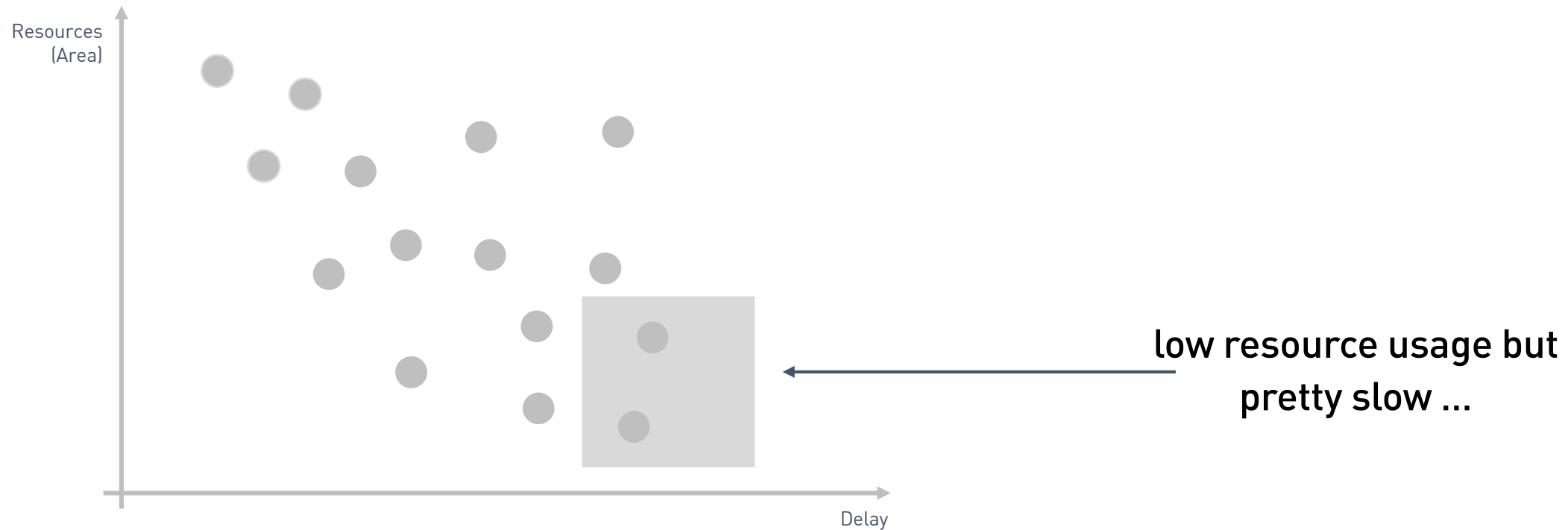
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FSoC WS22/23 in a Nutshell ...



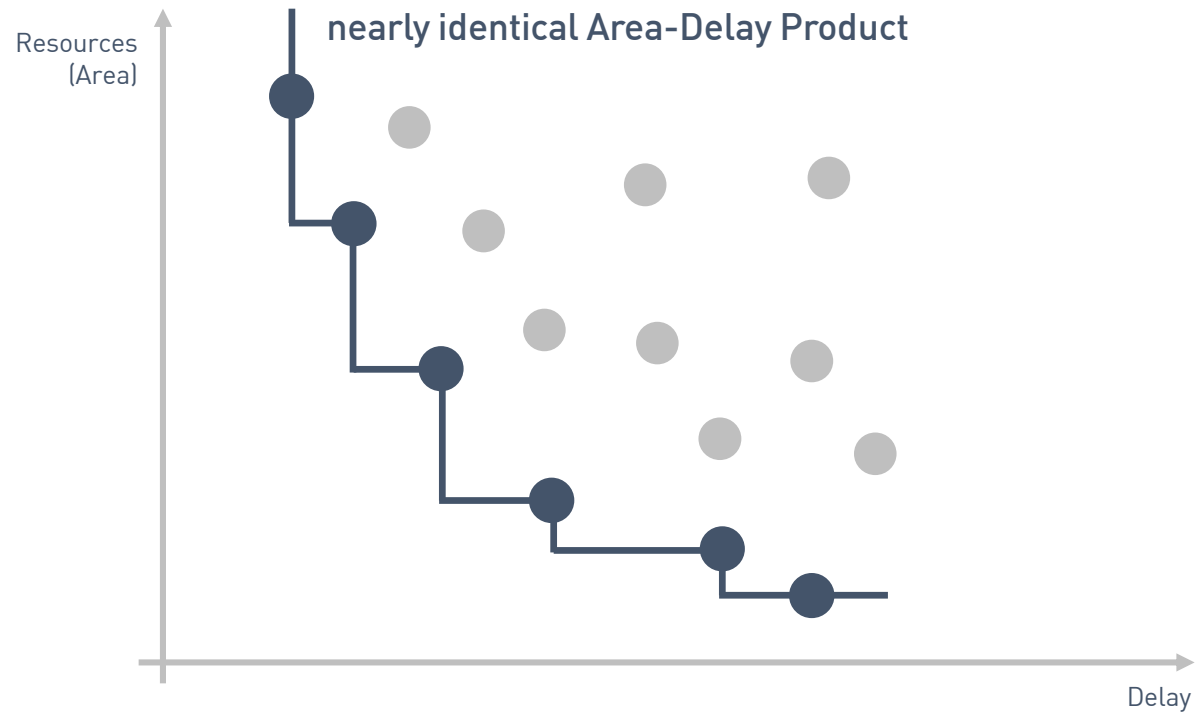
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- Which one to use ...?

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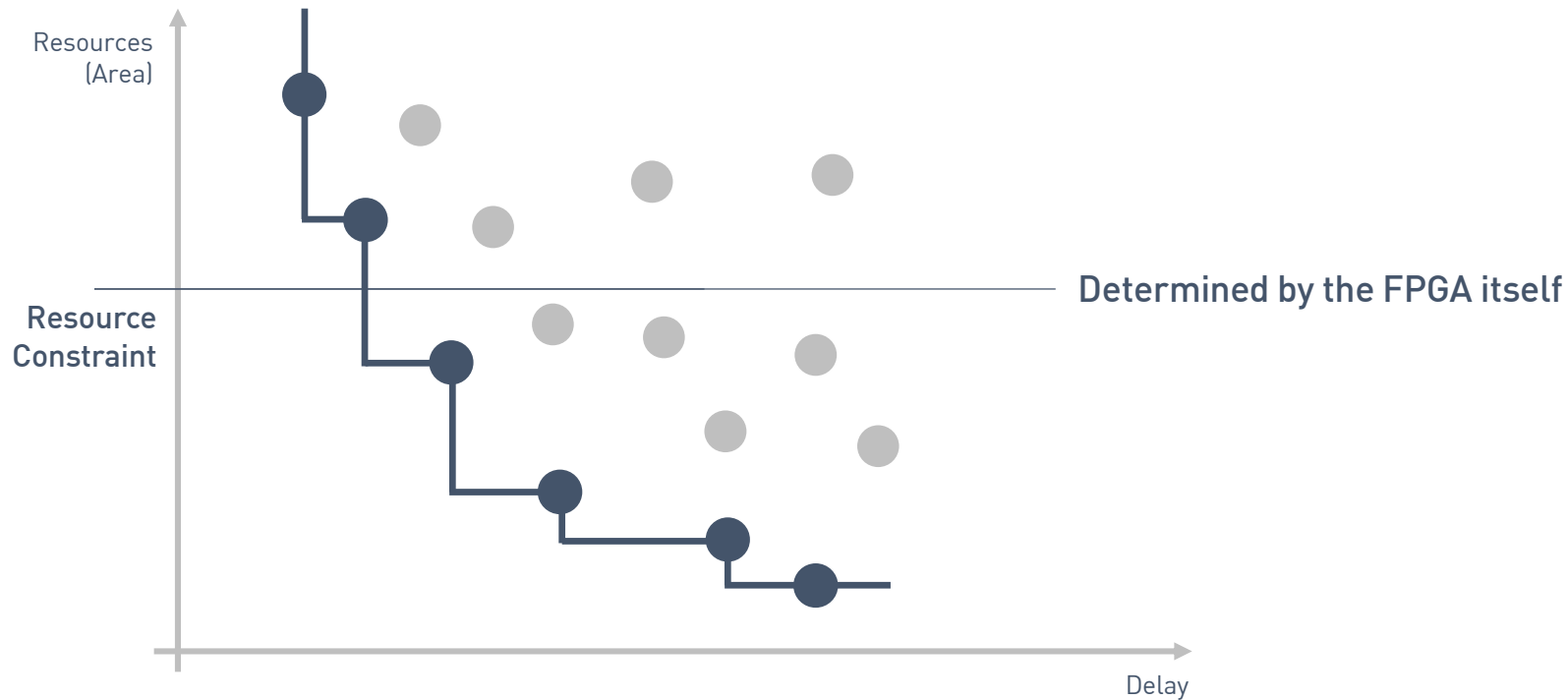
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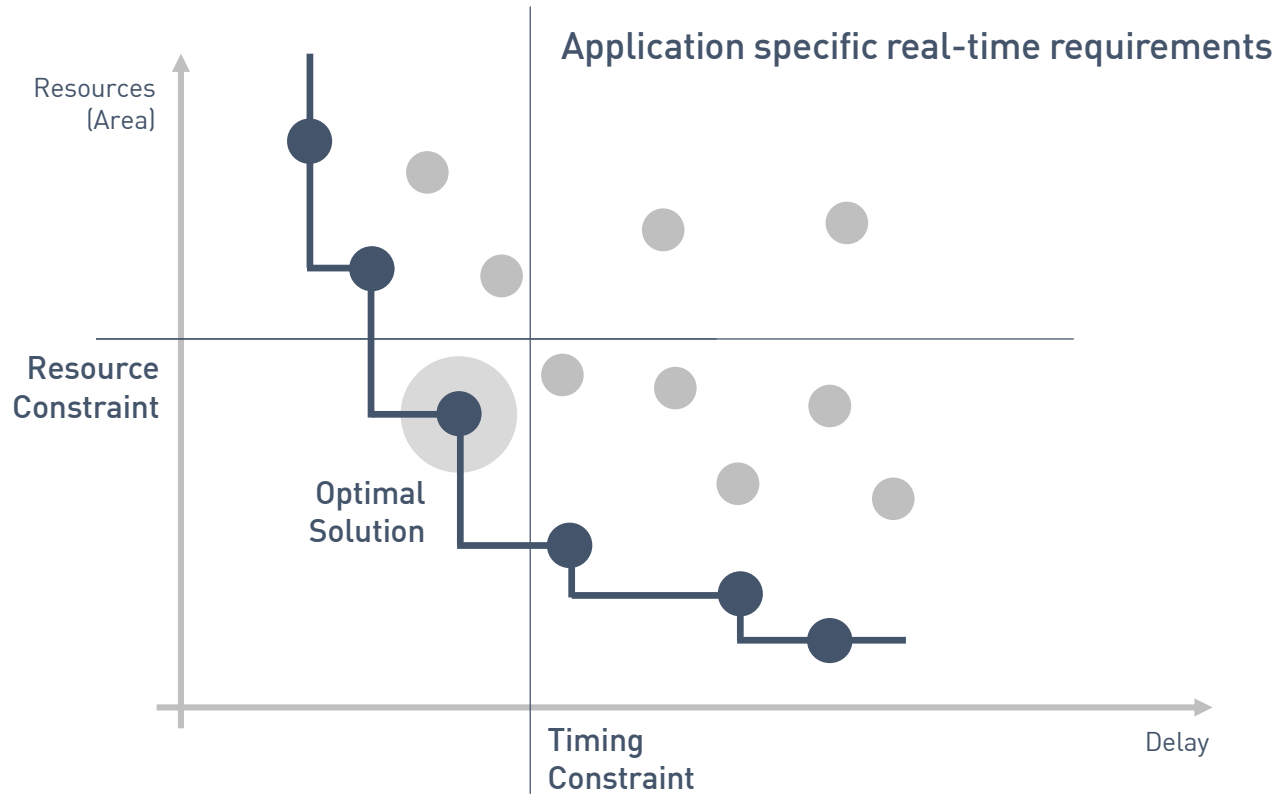
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FSoC WS22/23 in a Nutshell ...



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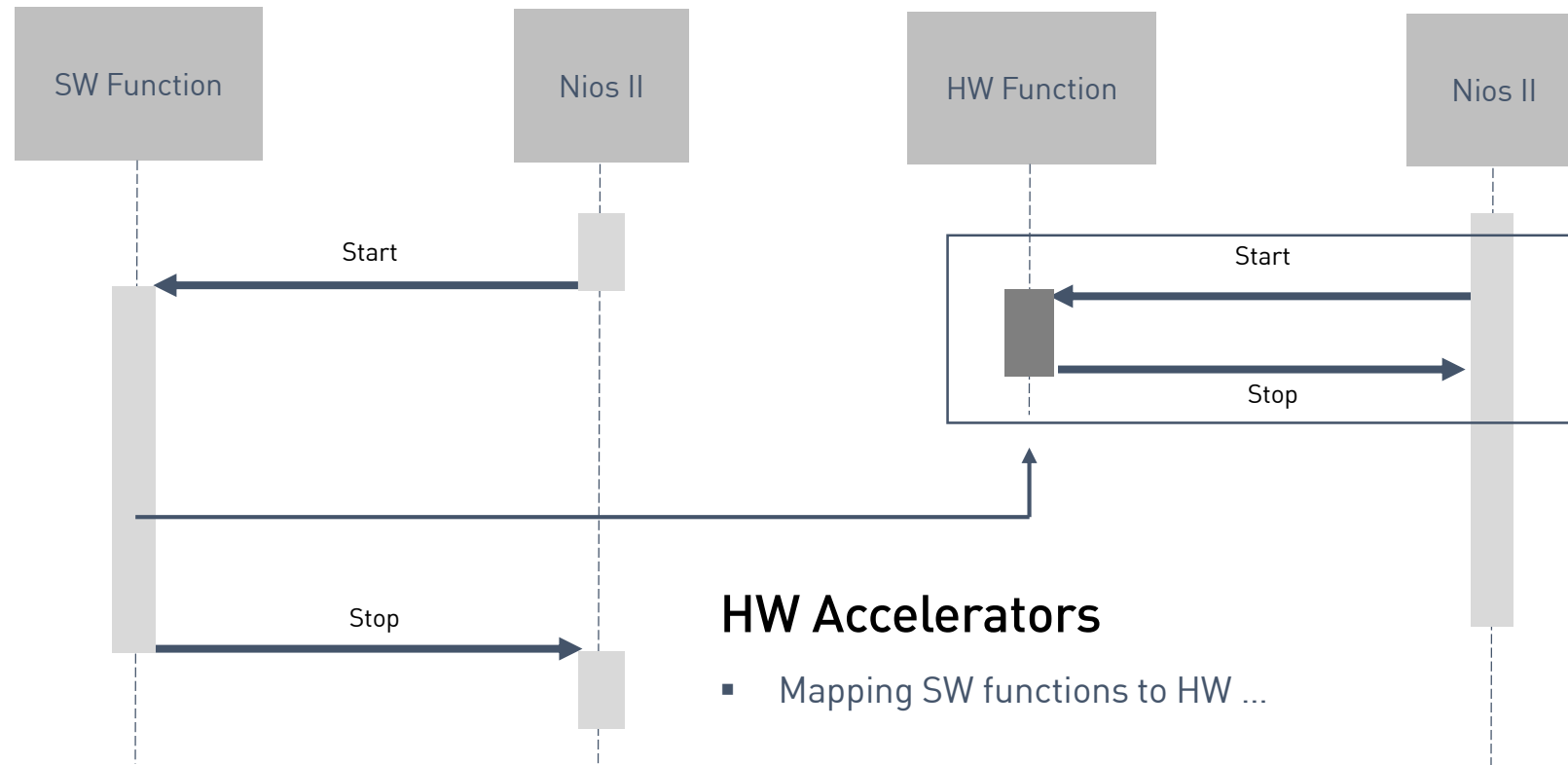
How to Speed up Software?

FPGA-based SoC Design

FSoC WS22/23 in a Nutshell ...

- How to speed up software ?

FSoC WS22/23 in a Nutshell ...



HW Accelerators

- Mapping SW functions to HW ...

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How to Design Custom Hardware Accelerators

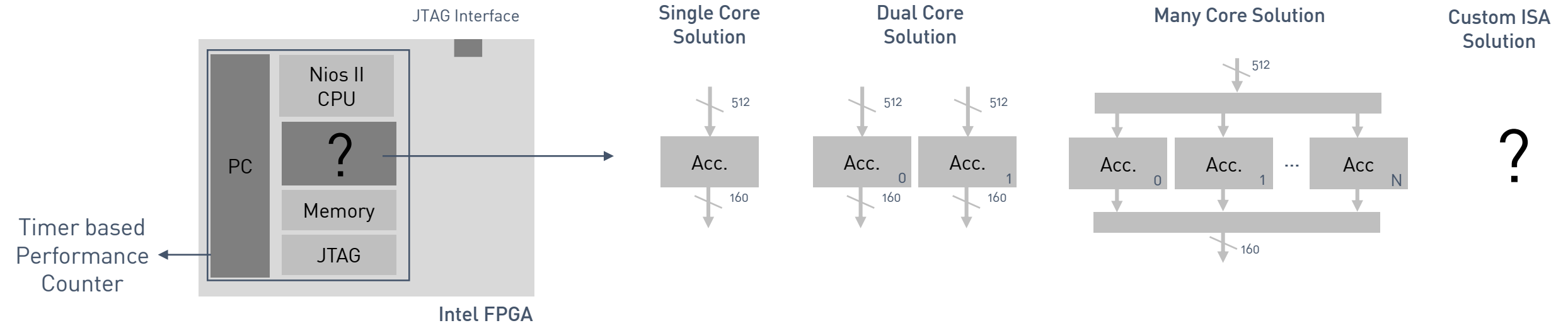
FPGA-based SoC Design

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FSoC WS22/23 in a Nutshell ...

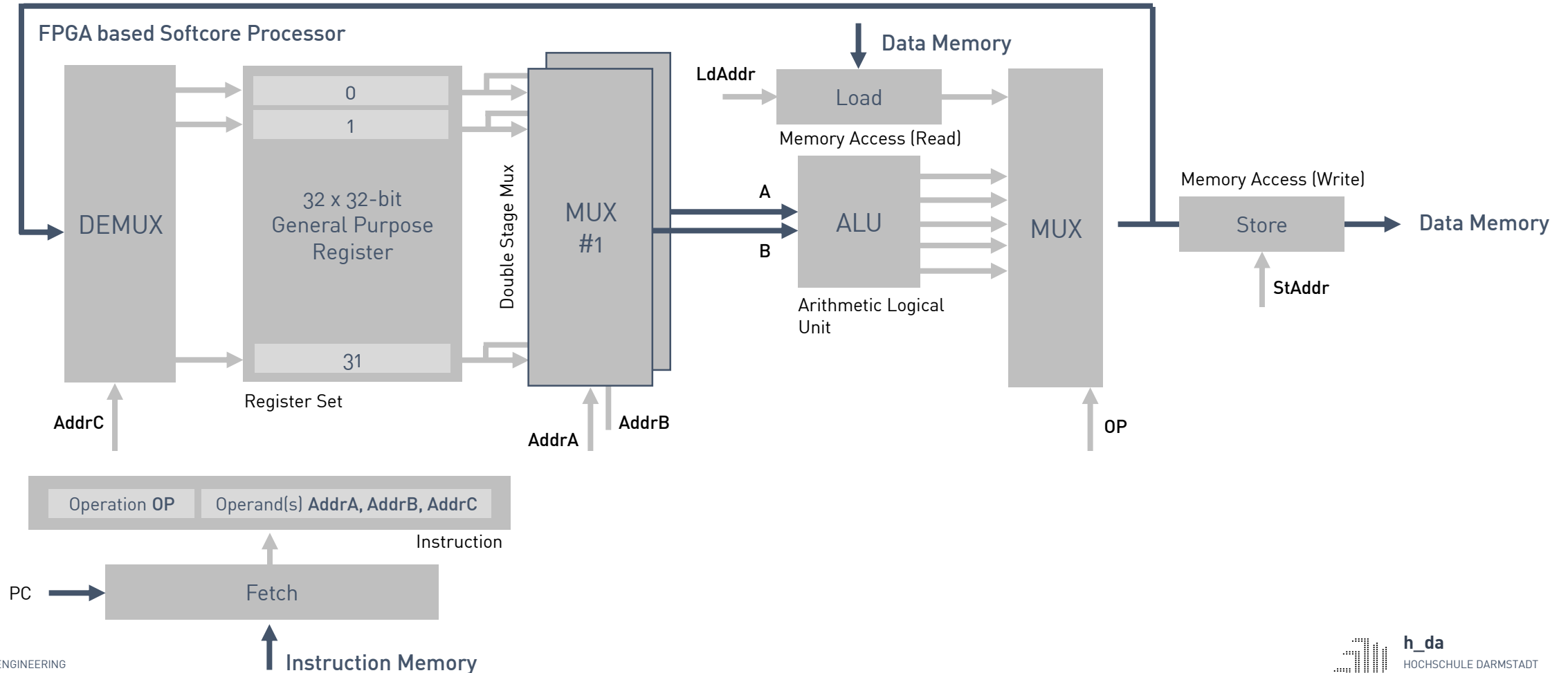
- How to design custom HW Accelerators ...



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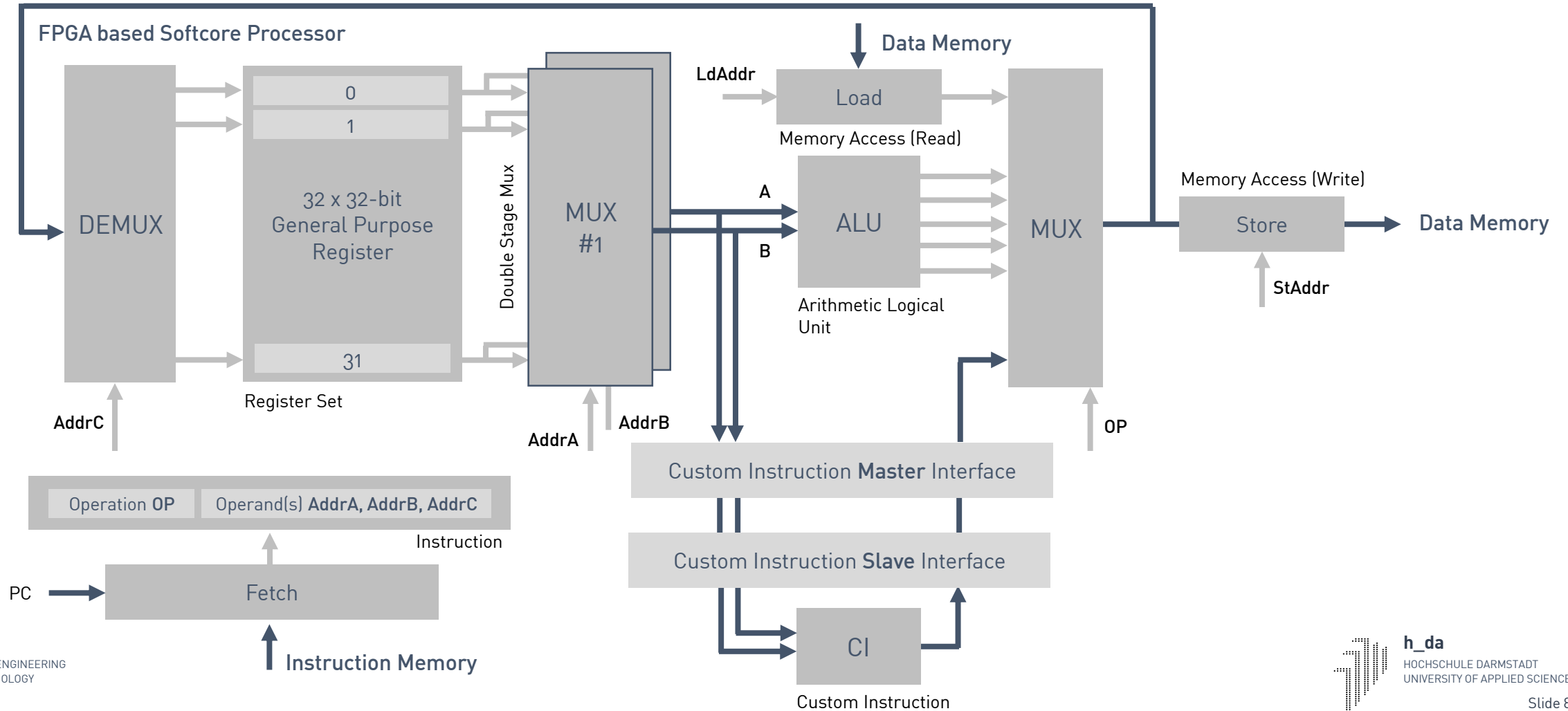
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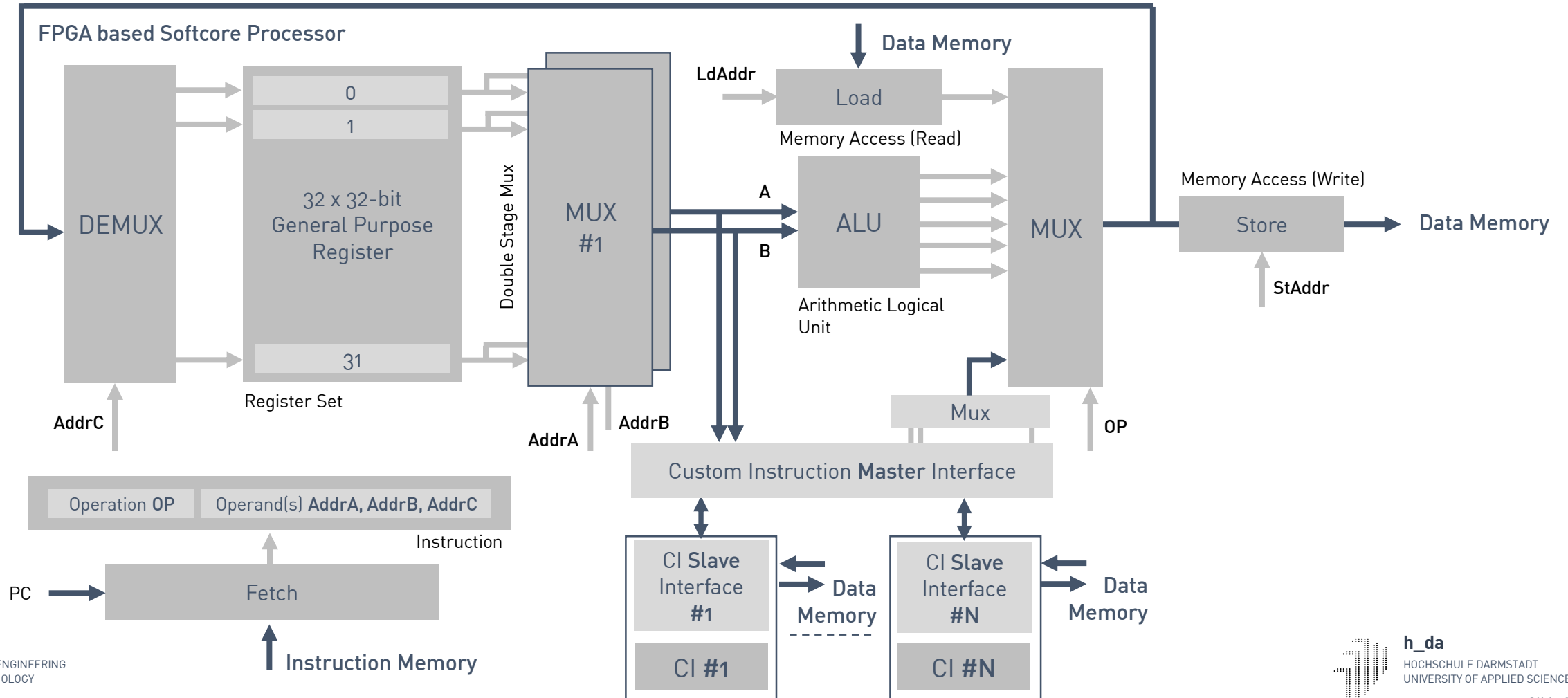
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Configuring the SoC FPGA to run the Linux OS

FPGA-based SoC Design

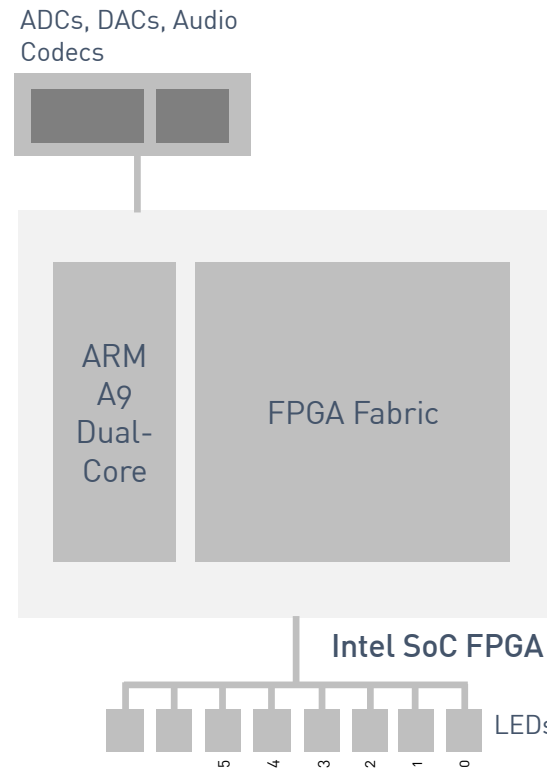
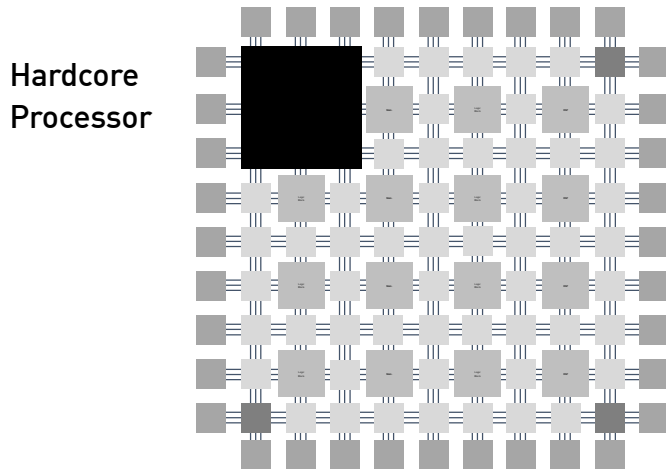
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FSoC WS22/23 in a Nutshell ...

- Hard Processor Systems using Embedded Linux

- How to design FPGA based SoC with integrated ARM A9 processor



The HPS integrates further peripherals:

- Several system control modules for clock and reset
- Debug support
- Timers (periodic and watchdog)
- Direct Memory Access Engine
- Ethernet Interface
- USB Interface
- UART
- CAN bus Controller
- SPI master and slave
- GPIO

Roadmap for next week

International Master of Science in Electrical Engineering

Prof. Dr. C. Jakob

University of Applied Sciences Darmstadt

h_da

Faculty of Electrical Engineering and Information Technology

fbeit

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Roadmap for next week

- Lecture #1 - Monday, 24th October, 10.15am
- Lecture #2 - Tuesday, 25th October, 12.30pm – Room tba.